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Attention, Herding, and Their Impact on Financial Markets

Evidence from Robinhood users, TAQ data, and U.S. Equities

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Abstract

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Keywords: retail investor attention; Robinhood; Google Trends; behavioral finance; empirical asset pricing

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I. Introduction

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II. Literature Review & Hypotheses

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III. Data & Samples Construction

3.1. Overview

This chapter describes the data architecture used to study retail investor attention, Robinhood herding, and the associated price dynamics. The empirical design combines four complementary data sources. First, the Robintrack dataset[1] provides security-level holdings of Robinhood users and therefore forms the backbone of the present attention-induced trading analysis. Second, CRSP supplies daily returns, closing and opening prices, market capitalization, as well as daily trading volume[2]. Third, TAQ data are used to construct an external measure of retail order flow based on the trade-classification procedure of Boehmer et al. (2021)[3]. Fourth, Google Trends and RavenPack[4] data are used to proxy, respectively, *direct* investor attention as Da et al. (2011, 2025), and the contemporaneous news or sentiment environment as De Giorgi et al. (2026).

Furthermore, due to data availability, the analysis is conducted on a sequence of nested samples rather than on a single balanced panel. The broadest sample is the *baseline trading sample*, constructed by merging the daily Robintrack variables with CRSP and TAQ by ticker and date. This sample is used to describe the evolution of Robinhood participation and validate Robintrack-based activity against TAQ retail order flow. The second sample is the *attention sample*, obtained by restricting the baseline sample to stock-day observations for which a Google Trends measure of search intensity can be reliably assigned. This restriction is non-trivial and economically important, because the Google Trends variable is the central outcome in the attention analysis and therefore becomes the main constraint on sample coverage. Finally, for specifications that condition on the information environment, I augment the attention sample with ticker-day measures derived from RavenPack. This distinction across samples is important for interpretation. The baseline sample is designed to preserve comparability with Barber et al. (2022). By contrast, the attention sample is tailored to the analysis of search activity and is necessarily narrower because exhaustive Google Trends coverage is difficult to obtain and sensitive to the precise construction of

[1] The Robintrack dataset is freely available at https://robintrack.net. This API exploited Robinhood's public API to record the number of users holding each stock at various times during the day.

[2] More precisely, I rely on the CRSP daily stock file, as made available by Wharton Research Data Services (WRDS).

[3] This supplementary dataset is incorporated to the analysis because it provides external validation of the Robintrack-based measures of retail trading activity, which is important given that Robinhood users represent only a subset of the overall retail investor population (Barber et al., 2022).

[4] RavenPack News Analytics is a data analytics provider in financial services and offers a comprehensive database of news and sentiment indicators. Inspired by the work of De Giorgi et al. (2026), I use RavenPack data to control for the contemporaneous news and sentiment environment. The RavenPack full equity dataset is proprietary and was accessed through WRDS.

the search variable[5]. In particular, recent evidence by deHaan et al. (2024) shows that conventional ticker-based Google search measures can contain substantial measurement error, implying that careful search-term design is essential when Google Trends is used as a proxy for investor attention. For that reason, the attention analysis is presented separately from the broader Robintrack-based descriptives, and the chapters distinguish clearly between the full trading universe and the subset of observations for which attention can be measured with sufficient precision.

3.2. Robinhood and Robintrack

3.2.1. Robinhood Markets Inc.

Robinhood Markets, Inc. is a U.S. financial services company headquartered in Menlo Park, California. Founded in 2013 by Vladimir Tenev and Baiju Bhatt, Robinhood states that its mission is "democratizing finance for all"[6]. Early in its development, the firm differentiated itself by offering commission-free trading[7] through a gamified mobile app and web interface. This model quickly gained traction, particularly among younger, less experienced, and smaller retail investors attracted by its low barriers to entry and ease of use. *The New York Times* reported that the average Robinhood user is 31 years old, has likely no prior trading experience[8], and, scaled by average account size, trades "nine times as many shares as E-Trade customers, and 40 times as many shares as Charles Schwab customers, in the most recent quarter"[9]. These observations corroborate Barber and Odean (2002, p. 791) who warn that "overconfident investors [young and inexperienced investors encouraged to trade by a gamified interface are likely to qualify] will trade too frequently", and earn suboptimal returns (Barber & Odean, 2001). The *The New York Times* also reports on the dangerous and hazardeous trading behaviors the gamified interface encourages[10]. Ozik et al. (2021, pp. 2357–2359) further show that Robinhood's user base, along with that of similar platforms, expanded sharply during the COVID-19 pandemic, as households were confined at home and experienced an increase in savings. In mid-2020, the company reported to have roughly 13 million users[11]. Taken together, these features make Robinhood

[5] Discussions with the authors of Barber et al. (2022) revealed that their solution to obtain Google Trends data for a large number of tickers through the unofficial R package `gtrendsR` was fastidious and involved a solution including IP addresses switching to avoid Google Trends' rate limits, as well as sleep intervals to avoid triggering Google's anti-bot mechanisms.

[6] https://robinhood.com/us/en/about-us/.

[7] Before its subsequent expansion into additional business lines, Robinhood's revenue model relied predominantly on payment for order flow (PFOF), under which brokers receive compensation from market makers for routing orders to them.

[8] Approximately one-half of users has no prior trading experience.

[9] https://www.nytimes.com/2020/07/08/technology/robinhood-risky-trading.html.

[10] https://www.nytimes.com/2021/01/30/business/robinhood-wall-street-gamestop.html.

[11] Over time, Robinhood also became associated with several controversies, including limited disclosure regarding its PFOF-based business model—as evidenced by Bloomberg, platform outages during periods

a particularly informative setting for the study of retail investor behavior.

3.2.2. *Robintrack Data*

To study the behavior of Robinhood users, I use the Robintrack dataset, which has become standard in the recent literature. Robintrack relied on Robinhood's public API to record the number of users, $N_{t,i}$, holding stock i at time t. Listing 3.1 reports a snapshot of the raw observations for Apple Inc. (AAPL).

Listing 3.1. Original Excerpt of Robintrack Observations for AAPL

```
timestamp,users_holding
"2018-05-02 04:53:46",150785
"2018-05-02 06:38:58",150785
"2018-05-03 00:35:25",145510
```

The dataset spans May 2, 2018 to August 13, 2020. Data collection ends on August 13, 2020, when Robinhood discontinued public access to the API used by Robintrack. The raw data are affected by four disruptions. As noted above, Robinhood experienced system outages during periods of unusually high trading activity on March 2, 2020 and March 9, 2020. In addition, the Robintrack collection script failed on August 9, 2018, January 24-29, 2019, and January 7-15, 2020. Following Welch (2022), I begin the baseline trading sample on June 1, 2018. The resulting baseline trading sample contains observations for 543 trading days and 8,597 usable tickers[12]. To aggregate the intraday observations to a daily frequency, I follow the procedure in Barber et al. (2022) and rely on Welch (2022) for several implementation details[13]. The main variables constructed from the Robintrack data are *users_close*, *users_last*, *userchg*, and *userratio*. The primary variable of interest is *users_close*, defined as the number of users holding a given stock after 2:00 p.m. ET but before the market close (4:00 p.m. ET). Regarding the particular timestamp of each observation, Barber et al. (2022) first emitted concerns about a delay between actual observation time and retrieval time. This delay was later confirmed to be 45 minutes by Ardia et al. (2023) who discussed with Casey Primovic, the developer of Robintrack. Therefore, for each timestamp $t_{i,k}$ in my dataset, I substract 45 minutes and convert it from

of elevated trading activity, alleged deficiencies in anti-money-laundering and cybersecurity controls, and, most prominently, restrictions on trading in several stocks during the GameStop short squeeze in January 2021. Had Robinhood not restricted trading in GameStop (GME) during the short squeeze, the firm may well have been pushed into bankruptcy because it could not have met the collateral requirements demanded by the National Securities Clearing Corporation (NSCC) https://fortune.com/2021/02/02/robinhood-gamestop-restricted-trading-meme-stocks-gme-amc-vlad-tenev-nscc/.

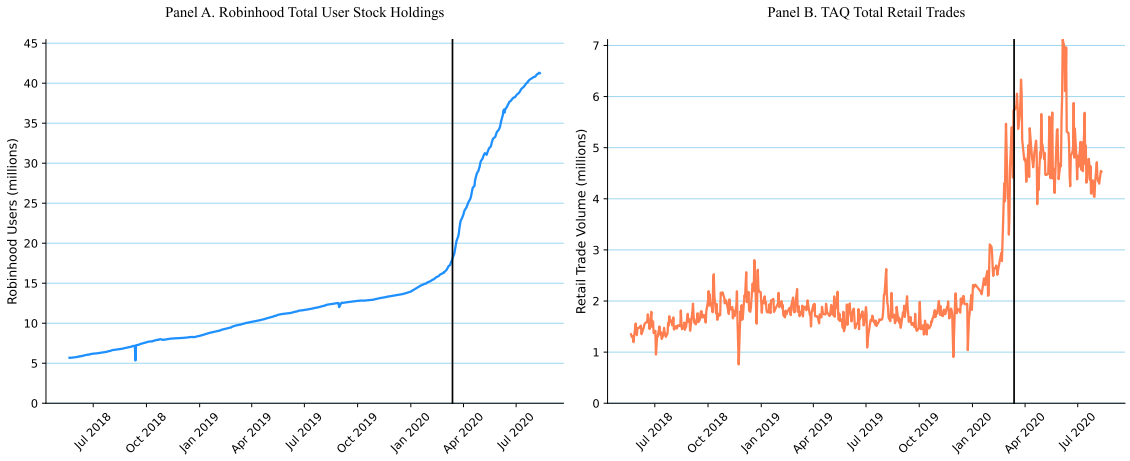
[12] To remain aligned with Barber et al. (2022), the baseline trading sample is not restricted to ordinary common shares (share code 10 or 11).

[13] For example, Welch (2022) notes that the Robintrack data contain four invalid tickers (_OUT, _PRN, MTL-, and PKD~), a detail not addressed in Barber et al. (2022).

UTC to ET to obtain the actual observation time. I then define *userchg* as the daily change in *users_close* and *userratio* as the ratio $users_close(t)$ / $users_close(t-1)$.

After constructing the daily variables from the Robintrack data, I merge them with the Center for Research in Security Prices (CRSP) daily stock file by ticker and date. To adjust CRSP returns for delisting events, I follow the procedure in Bali et al. (2016). I then augment the sample with Trade and Quote (TAQ) data from Wharton Research Data Services (WRDS) to obtain an external validity measure of retail trading activity, namely retail buy and sell executions. As Boehmer et al. (2021) note, the emergence of algorithmic and high-frequency trading made traditional trade-size-based classifications of retail and institutional trading increasingly unreliable, because large institutional orders were often split into many small trades. To address this problem, Boehmer et al. (2021, pp. 2254–2256) develop a classification algorithm grounded in the regulatory structure of U.S. equity markets. Their approach exploits the fact that retail marketable orders are typically executed off-exchange by wholesalers or internalizers and frequently receive small subpenny price improvement, whereas institutional orders generally do not. Accordingly, they focus on off-exchange TAQ trades reported to the FINRA TRF (exchange code "D") and classify transactions using the fractional component of the trade price relative to the nearest cent. Trades reported just above a round penny are classified as retail sells, whereas trades reported just below a round penny are classified as retail buys. More precisely, for TRF-reported trades, transactions with a fractional cent component in the interval $(0, 0.4)$ are coded as retail sells, while those in $(0.6, 1)$ are coded as retail buys; trades at round-penny prices or near half-penny prices are excluded.

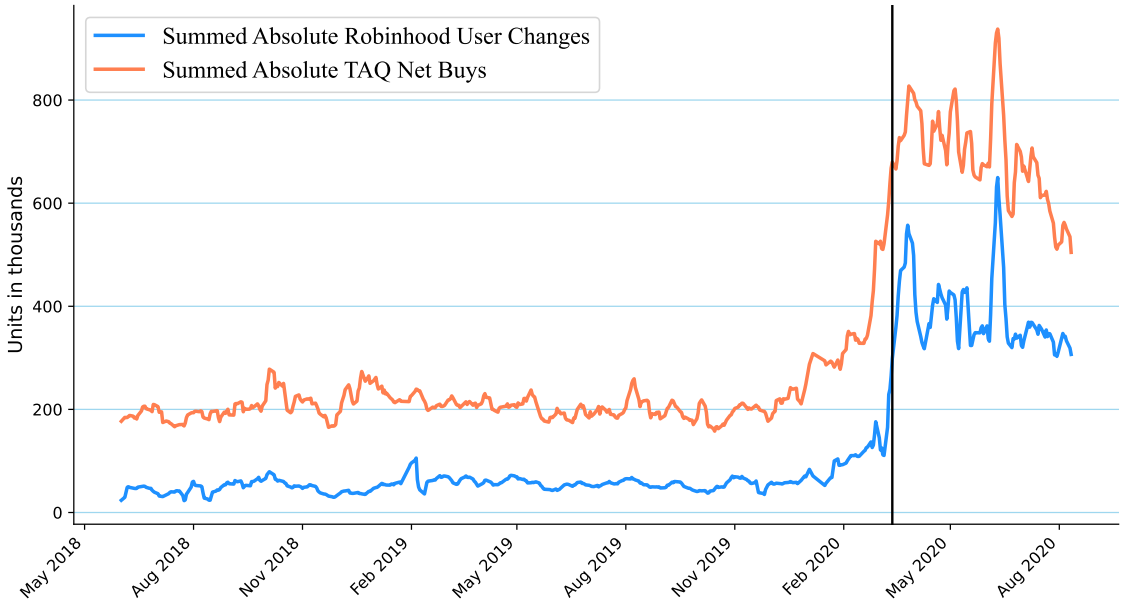
Figure 1. Robinhood Users and TAQ Retail Volume. Panel A plots the total number of Robinhood users holding stocks at the market close (*users_close*) over the sample period. Panel B plots the daily total number of retail trades executed off-exchange, as classified by Boehmer et al. (2021). The red vertical line indicates the date when COVID national emergency was declared in the U.S. (March 13, 2020).



From Figure 1, we see that the observation of Ozik et al. (2021) is indeed correct; both

the number of Robinhood users total stock holdings and the total number of TAQ retail trades increased sharply during the COVID-19 pandemic. Of course, the two measures are partially overlapping, as the Robinhood user base is a subset of the overall retail investor population.

Figure 2. Absolute Changes of Robinhood Users Stock Holdings and TAQ Retail Net Buys. This figure plots the evolution over the period of the 5-day moving average of the absolute daily change in the number of Robinhood users holding stocks at the market close (*userchg*), as well as the 5-day moving average of the absolute daily net number TAQ retail buys. The red vertical line also indicates the date when COVID national emergency was declared in the U.S. (March 13, 2020).



In Figure 2, I further plot—inspired by Barber et al. (2022)—the 5-day moving average of the absolute daily change in the number of Robinhood users holding stocks at the market close (green line), as well as the 5-day moving average of the absolute daily net number TAQ retail buys (blue line). Again, this suggests that both measures follow similar patterns, with a pronounced increase in activity during the COVID-19 pandemic[14]. Unfortunately, the Robintrack data end in August 2020, so we cannot observe whether the elevated levels of retail trading activity persisted beyond the pandemic[15].

Finally, Table 1 reports summary statistics for the baseline trading sample. The average number of Robinhood users holding a given stock is about 2,141, but the distribution is highly skewed: the median is 173, whereas the maximum is close to 1 million for the most widely held stock[16]. This pattern suggests that Robinhood holdings, and likely retail

[14] No formal statistical test is conducted to compare the two time series, but Barber et al. (2022) provide empirical evidence supporting that Robinhood users proxy relatively well for the overall retail investor population.

[15] The recent work of Bryzgalova et al. (2023) in retail trading of options suggests that the elevated levels were indeed record highs but the increased activity could persist for a while.

[16] The five stocks with the highest average number of Robinhood holders are Aurora Cannabis Inc. (ACB),

Table 1. Summary Statistics of the Baseline Trading Sample

Panel A presents summary statistics across stock-day observations. Panel B sums variables by day and then averages across days. The variables are *users_close* (last observed user count for a stock prior to the 4 p.m. ET close), *users_last* (last user count of the day), *userchg* (daily change in *users_close*), *userratio* ($users_close_t$ / $users_close_{t-1}$), *prc* (closing price), *size* (market cap in millions), *ret* (daily return), *daily_buys* (number of TAQ daily retail buys), *daily_sells* (number of TAQ daily retail sells), *net_buys* (*daily_buys* − *daily_sells*), *taq_retimb* (*net_buys*/(*daily_buys*+*daily_sells*)), and #stocks (number of stocks with users reported on Robintrack). This table follows the design of Barber et al. (2022, p. 3151).

Variable	*N*	Mean	SD	min	p25	p50	p75	max
Panel A: Stock-Day Observations								
users_close	3,755,888	2,142.95	15,820.03	0.00	40.00	173.00	714.00	990,444.00
users_last	3,841,684	2,140.93	15,830.70	0.00	40.00	173.00	713.00	990,587.00
userchg	3,678,526	9.72	383.71	−548,766.00	−1.00	0.00	1.00	85,193.00
userratio	3,639,029	1.01	1.18	0.00	1.00	1.00	1.00	2,087.00
prc	3,841,684	41.07	91.59	0.04	10.46	23.72	46.67	4,699.00
size ($mil.)	3,766,791	5,603.11	30,996.71	0.03	93.41	442.72	2,249.69	1,966,963.00
ret (%)	3,823,449	0.04	4.29	−100.00	−0.98	0.02	0.99	1,025.18
daily_buys	3,659,438	199.32	1,105.93	0.00	8.00	33.00	116.00	185,252.00
daily_sells	3,659,438	177.84	871.19	0.00	8.00	33.00	114.00	112,743.00
net_buys	3,659,438	21.48	365.31	−30,215.00	−7.00	0.00	10.00	86,325.00
taq_retimb	3,659,438	0.01	0.35	−1.00	−0.14	0.00	0.16	1.00
Panel B: Daily Observations (Summed Variable Averaged across Days)								
#stocks	554	6,706.95	702.91	180.00	6,197.50	6,634.00	7,458.75	7,614.00
users_close (mil.)	554	14.53	9.70	0.14	8.12	11.58	14.86	41.32
users_last (mil.)	554	14.54	9.71	0.14	8.12	11.58	14.87	41.33
userchg (000)	554	64.55	110.49	−185.82	12.92	22.09	51.80	817.19
daily_buys (000)	554	1,285.74	751.43	9.84	831.40	932.08	1,308.58	3,972.02
daily_sells (000)	554	1,146.70	598.58	9.30	783.01	879.50	1,193.29	3,200.01
net_buys (000)	554	139.04	172.09	−68.88	38.73	63.54	135.39	1,007.41

attention, are concentrated in a relatively small set of stocks, consistent with prior evidence in Barber et al. (2022), Münster et al. (2024), and Welch (2022). Table 1 further shows that the daily change in the number of Robinhood users holding a stock (*userchg*) is positive on average but economically small, with a mean of 9.72 users and a median of zero. This is consistent with the evidence in Figure 2 and with the positive mean of TAQ retail *net_buys*.

3.3. Google Trends

3.3.1. Conceptual Motivation

In the attention-induced trading literature, we often measure investor attention with *indirect* proxies, such as extreme abnormal returns (Barber & Odean, 2008), trading volume (Barber & Odean, 2008; Hou et al., 2009), news and headlines (Ballinari et al., 2022; Barber & Odean, 2008; Barber et al., 2022), or price thresholds (Kupfer & Schmidt, 2021). However, as Da et al. (2011, p. 1462) note, these "indirect" proxies do not capture investor attention but rather potential sources of attention—a "news article in the Wall Street Journal does not guarantee attention unless investors actually read it".

Fortunately, over the past decade, the availability of new data sources has enabled the use of *direct* measures of investor attention, such as social media activity (Audrino et al.,

Ford Motor Company (F), General Electric Company (GE), Apple Inc. (AAPL), and Microsoft Corporation (MSFT).

2024; De Giorgi et al., 2026; Fedyk, 2021) and Google search activity (Da et al., 2011, 2025; deHaan et al., 2024). These proxies are referred to as "direct" because when someone searches for a stock ticker on Google, or posts about it on Reddit[17], it is more likely that they are actually paying attention to the stock, as opposed to simply being exposed to a news article or an extreme return. Hence, in line with Da et al. (2011, 2025), I use Google Trends search volume index (SVI) as the main proxy for investor attention.

3.3.2. Variable Construction

Google provides a publicly available tool, Google Trends[18], to track the search volume intensity of specific keywords over time. The search volume index (SVI) provided by Google Trends is a normalized measure of the search interest for a given keyword[19], scaled from 0 to 100, where 100 represents the peak search interest for that keyword during the specified time period and geographic region. For an illustration of the raw SVI, the reader can refer to Figure 10 in the Appendix. Based on the recommendations in deHaan et al. (2024), I rely on their construction of SVI variables, which are designed to mitigate measurement error and noise searches[20]. Contrary to Barber et al. (2022) and Da et al. (2011) who rely on weekly Google Trends data, the dataset I use to build myattention sample relies on daily SVI data, as in Da et al. (2025). Due to the difficulty of scaling Google Trends data across a large number of tickers, SVI time series are constructed only for stocks in the Russell 3000 index[21], also following Da et al. (2011). Once I obtain the daily SVI for each stock—denoted by *svi*—I merge them with the baseline trading sample by ticker and date to construct the attention sample. This procedure results in an attention sample containing over 1 million stock-day observations for 1,979 unique tickers. Finally, I also construct the variable $asvi_{i,t-1}$, inspired by the approach in Barber et al. (2022) and outlined in Equation 3.1, to capture the lagged abnormal search interest for a stock i on the previous day $t-1$.

[17] De Giorgi et al. (2026) uses text processing techniques to quantify the content of the subreddit `r/wallstreetbets`.

[18] https://trends.google.com/trends/

[19] Google samples a random subset of all searches and normalizes the search volume for a keyword by the total number of searches in the specified time period and geographic region. The resulting SVI is then scaled from 0 to 100, where 100 represents the peak search interest for that keyword during the specified time period and geographic region. See https://newsinitiative.withgoogle.com/resources/trainings/basics-of-google-trends/.

[20] For instance, Da et al. (2011) notices that the ticker "CAT" is also a common English word, which can lead to noisy search data. To address this issue, deHaan et al. (2024) constructs a more precise search term for CAT by using the syntax "CAT stock" in Google Trends. This approach helps to filter out irrelevant searches and provides a more accurate measure of investor attention for the stock. The datasets of deHaan et al. (2024) are publicly available on their `GitHub` repository at https://github.com/robinlitjens/GoogleTickerStock-SVI/tree/main.

[21] The Russell 3000 index is a widely followed benchmark that includes the 3,000 largest U.S. stocks by market capitalization, representing approximately 98% of the investable U.S. equity market. See https://www.ftserussell.com/products/indices/russell-3000-index.

$$asvi_{i,t-1} = \ln\left(1 + \frac{svi_{i,t-1}}{\frac{1}{20}\sum_{\tau=2}^{21} svi_{i,t-\tau}}\right) \tag{3.1}$$

Table 2 reports summary statistics for the attention sample similar to Table 1 and allow for comparison between the two samples.

Table 2. Summary Statistics of the Attention Sample

Panel A presents summary statistics across stock-day observations. Panel B sums variables by day and then averages across days. The variables are *users_close* (last observed user count for a stock prior to the 4 p.m. ET close), *users_last* (last user count of the day), *userchg* (daily change in *users_close*), *userratio* ($users_close_t$ / $users_close_{t-1}$), *prc* (closing price), *size* (market capitalization), *ret* (daily return), *daily_buys* (number of TAQ daily retail buys), *daily_sells* (number of TAQ daily retail sells), *net_buys* (*daily_buys* − *daily_sells*), *taq_retimb* (*net_buys*/(*daily_buys*+*daily_sells*)), #stocks (number of stocks with users reported on Robintrack), *svi* (Google Trends search volume index), and *asvi* (abnormal search volume index defined in Equation 3.1).

Variable	*N*	Mean	SD	min	p25	p50	p75	max
Panel A: Stock-Day Observations								
users_close	1,037,118	3,794.27	23,947.79	1.00	96.00	306.00	1,193.00	943,931.00
users_last	1,037,118	3,797.05	23,967.66	0.00	96.00	306.00	1,193.00	944,388.00
userchg	1,015,432	17.02	334.42	−17,015.00	−2.00	0.00	2.00	51,733.00
userratio	1,015,406	1.01	0.27	0.01	1.00	1.00	1.01	136.74
prc	1,037,118	66.83	151.54	0.09	16.20	36.80	75.16	4,699.00
size ($mil.)	1,037,095	14,514.38	55,373.50	1.17	671.20	2,324.30	8,479.41	1,948,022.00
ret (%)	1,037,095	0.03	3.70	−100.00	−1.16	0.04	1.19	585.00
daily_buys	1,031,902	336.89	1,504.90	0.00	23.00	76.00	231.00	185,252.00
daily_sells	1,031,902	308.70	1,217.37	0.00	24.00	77.00	229.00	112,743.00
net_buys	1,031,902	28.18	453.17	−10,131.00	−12.00	0.00	15.00	86,325.00
taq_retimb	1,031,902	0.00	0.24	−1.00	−0.10	0.00	0.10	1.00
asvi	967,061	0.45	0.72	0.00	0.00	0.00	0.82	5.12
Panel B: Daily Observations (Summed Variables Averaged across Days)								
#stocks	554	1,872.05	116.20	11.00	1,863.00	1,878.00	1,896.00	1,917.00
users_close (mil.)	554	7.10	4.80	0.00	4.27	5.42	6.43	20.32
users_last (mil.)	554	7.11	4.80	0.00	4.27	5.42	6.43	20.32
userchg (000)	554	31.19	63.21	−64.71	3.98	8.58	19.66	467.90
daily_buys (000)	554	627.50	358.80	0.35	416.36	464.68	619.32	2,126.34
daily_sells (000)	554	575.00	280.68	0.33	407.62	455.30	598.06	1,728.91
net_buys (000)	554	52.50	91.13	−54.64	3.08	16.93	49.38	579.26

Compared with the baseline trading sample in Table 1, the attention sample in Table 2 is tilted toward larger and more actively followed stocks, consistent with its restriction to Russell 3000 firms. Relative to the baseline sample, stocks in the attention sample have substantially higher market capitalization (mean *size* of $14,514.38 million versus $5,603.11 million), are held by more Robinhood users (mean *users_close* of 3,794.27 versus 2,142.95), and exhibit greater retail trading activity (mean *daily_buys* of 21.48 versus 28.18). In addition, the distributions of *svi* and *asvi* are highly right-skewed, indicating that, on most stock-days, search activity is limited, whereas a small subset of stocks attracts disproportionately high investor attention[22].

[22] In the full attention sample, 34.14% of stock-day observations have $svi > 0$, 16.93% have $svi > 50$, and only 3.14% have $svi = 100$.

3.4. RavenPack News Analytics

To control for the potential confounding effect of news coverage and overall investor sentiment, I construct some controls following Audrino et al. (2020) and De Giorgi et al. (2026). I augment the attention sample with news analytics data from RavenPack News Analytics, a leading provider of structured news data and sentiment analysis[23]. Inspired by the approach in De Giorgi et al. (2026), I build five control variables to capture the intensity and sentiment of news coverage for each stock-day observation. First, I construct *num_news* as the total number of press releases and news articles related to a stock on a given day and *num_news_relevant* as the number of news items for which RavenPack assigns a relevance score of at least 75 out of 100[24]. Subsequently, I construct an Event Sentiment Score *ess*, a Composite Sentiment Score *css*, and an Analyst Recommendation Change indicator *anl_chg*. These measures are aggregated over market hours to obtain stock-day level variables, which are then merged with the attention sample by ticker and date. A detailed description of the RavenPack News Analytics dataset and the construction of these variables can be found in Appendix A.1.2. Importantly, sentiment is controlled for because, as evidenced by Cookson et al. (2024), sentiment and attention contain different return-relevant information and might influence retail trading in opposite directions.

3.5. Herding Events

Table 1 and Table 3 demonstrated that daily user changes are generally small, but as presented by Barber et al. (2022, p. 3150), the data sample contains several instances of "extreme events" in which the number of users holding a stock changes drastically. These events are of particular interest, as they likely reflect episodes of intense retail attention and herding. To identify such events, I follow the procedure in Barber et al. (2022) and apply the following filters to the attention sample. First, I restrict the sample to stocks with at least 100 users on the previous day, to avoid extreme percentage changes driven by small denominators. Second, I focus on days with an increase in the number of users (i.e., $userratio > 1$). Once these filters are applied, I proceed to sort the stock day observations based on *userratio* at time t and flag the top 0.5% of the distribution as herding events; denoted by the indicator *rh_herd*. The resulting sample contains 1,997 herding events for 864 unique tickers.

Table 3 reports summary statistics for the herding events identified in the sample. During these episodes, the average daily increase in the number of Robinhood users holding a stock

[23] Specifically, I use the RavenPack News Analytics dataset "full equity", as licensed by WRDS.

[24] In their user guide, RavenPack defines the relevance score as an integer score between 0-100 indicating how strongly related the mention of an entity is to the underlying news story, with higher values indicating greater relevance. For any news story that mentions an entity, RavenPack provides a relevance score. A score of 0 means the entity was passively mentioned while a score of 100 means the entity was prominent in the news story. Values above 75 are considered significantly relevant.

Table 3. Summary Statistics for Robinhood Herding Events

This table reports summary statistics for stock-days associated with herding events, defined as securities in the top 0.5% of positive *userchg* ratio on day t and with at least 100 users on day $t-1$. The variables are *users_close* (last observed user count for a stock prior to the 4 p.m. ET close), *users_last* (last user count of the day), *userchg* (daily change in *users_close*), *userratio* ($users_close_t$ / $users_close_{t-1}$), *prc* (closing price), *size* (market capitalization in millions of dollars), *ret* (daily return), *daily_buys* (number of TAQ daily retail buys), *daily_sells* (number of TAQ daily retail sells), *net_buys* (*daily_buys* − *daily_sells*), and *taq_retimb* (*net_buys*/(*daily_buys*+*daily_sells*)).

Variable	*N*	Mean	SD	min	p25	p50	p75	max
users_close	1,997	2,802.09	10,467.34	106.00	289.00	632.00	1,735.00	168,920.00
users_last	1,997	2,890.63	10,760.06	106.00	294.00	638.00	1,786.00	176,399.00
userchg	1,997	866.17	3,220.94	6.00	65.00	174.00	542.00	49,277.00
userratio	1,997	1.58	0.85	1.04	1.21	1.35	1.62	13.71
prc	1,997	41.51	110.17	0.23	7.30	19.97	44.37	2,990.00
size ($mil.)	1,997	5,077.66	17,919.27	2.75	425.19	1,075.23	3,408.15	425,082.69
ret (%)	1,997	2.66	25.14	−86.67	−9.93	1.02	10.91	281.27
daily_buys	1,996	2,522.61	8,411.31	1.00	231.00	606.50	1,752.75	185,252.00
daily_sells	1,996	1,896.76	5,144.24	0.00	200.00	529.00	1,532.00	98,927.00
net_buys	1,996	625.84	3,764.90	−5,413.00	−12.00	47.00	248.00	86,325.00
taq_retimb	1,996	0.08	0.17	−0.69	−0.02	0.07	0.18	1.00
svi	1,938	39.19	39.43	0.00	0.00	32.00	76.00	100.00
asvi	1,896	0.62	0.80	0.00	0.00	0.00	1.18	4.04

is about 866, consistent with the buy imbalance observed in the TAQ measure of retail order flow (*taq_retimb*). In addition, the average herding event is associated with a positive daily return of 2.66% but we also observe extreme negative returns; aligned with the view that retail attention is drawn to stocks experiencing large price moves (Barber & Odean, 2008)[25]. We also observe that stocks associated with herding events are substantially smaller than those in the full attention sample, with mean market capitalization of $5,077.66 million versus $13,905.87 million, suggesting that retail attention effects are more concentrated in small-cap stocks[26]. Finally, the average *svi* and *asvi* during herding events are 39.19 and 0.62, respectively, which are substantially higher than the average values in the full attention sample (19.24 and 0.45, respectively), insinuating that herding events could be associated with heightened investor attention as measured by Google search activity.

[25] Robinhood reinforces this pattern by displaying the "Top Movers" on the app's home screen, which highlights stocks with the largest price changes and thereby attracts retail attention to them.

[26] Such a pattern is consistent with De Giorgi et al. (2026, p. 5) stating that "social media attention mobilizes Robinhood users most strongly towards either niche small-caps, perceived as opportunities, or highly visible large-caps, associated with familiarity". Since my attention sample is restricted to Russell 3000 firms and therefore might exclude microcap stocks, I identify less extreme returns than those observed in Barber et al. (2022), who include microcap stocks in their sample.

IV. Attention, Stock Selection, & Herding Predictability

This chapter examines how limited investor attention is translated into retail investors' stock-selection decisions and into the predictability of discrete herding episodes. Motivated by the attention-driven buying hypothesis of Barber and Odean (2008) and Barber et al. (2022)—which emphasizes that attention matters more for purchases, where investors must search across thousands of candidates, than for sales, which are largely confined to previously held positions—I first document that retail trading is highly concentrated in a small subset of tickers, especially on the buy side and especially among Robinhood users. I then estimate stock and date fixed-effects predictive regressions that relate next-day Robinhood user flows to lagged abnormal search intensity and other salience proxies such as extreme returns, abnormal volume, and news coverage. Finally, I extend the analysis to linear probability models (LPMs) of herding and show that the attention–herding relation is most pronounced among smaller-cap stocks, consistent with previous findings (Da et al., 2011; De Giorgi et al., 2026; Fedyk, 2021; Garcia, 2026).

4.1. Concentration of Retail Trading

When retail investors decide whether to buy or sell a stock, they face fundamentally different choice problems once one accounts for the size of the relevant stock universe. Because short selling is typically unavailable to retail investors, sales are largely confined to the subset of stocks already held in the portfolio, so that the corresponding consideration set is extremely narrow.[1] By contrast, when choosing what to buy, they must search across thousands of potential candidates. This creates a severe search problem, particularly given retail investors' limited financial literacy (Lusardi, 2015) and limited attention (Kahneman, 2011). In this setting, the literature shows that retail investors tend to restrict their buying consideration set to attention-grabbing stocks, that is, stocks made salient by news coverage, social media, or other channels (Barber & Odean, 2008; Seasholes & Wu, 2007). If retail investors actively trade only a small subset of salient stocks, one should observe a high degree of concentration in retail trading activity, especially on the buy side. This section therefore examines and compares the concentration of Robinhood and TAQ retail trading by following the approach of Barber et al. (2022).

To assess the concentration of retail trading activity, I construct daily rank-share series separately for Robinhood user changes and TAQ retail net buying. For each stock-day, Robinhood net flow is measured by the daily change in the number of Robinhood users holding the stock ($userchg_{i,t}$), while TAQ net flow is measured by retail net buys

[1] On average, retail investors hold between three and four stocks in their portfolio (Seasholes & Wu, 2007).

($net_buys_{i,t}$). I then treat purchases and sales separately. On the buy side, I retain only stock-days with strictly positive net flow and interpret the flow itself as the amount bought. On the sell side, I retain only stock-days with strictly negative net flow and convert these values into positive magnitudes so that they can be interpreted as the amount sold[2]. For each trading day and for each series in turn, I rank stocks in descending order of the relevant flow magnitude and compute, for every rank, the share of aggregate daily net buying or net selling accounted for by that stock. I retain the top ten ranks and discard lower-ranked observations, so that each series measures the fraction of total daily activity captured by the ten largest stock-level flows. Following this procedure, Figure 3 reports, for each rank, the time-series average of these daily shares, expressed in percentage terms. Thus, the first bar corresponds to the average fraction of daily net buying or selling attributable to the single most heavily traded stock on a given day, the second bar to the second most heavily traded stock, and so forth through rank ten[3].

Figure 3. Concentration of Retail Buying and Selling. This figure reports the average fraction of daily net buying and net selling attributable to the ten most heavily traded stocks on Robinhood and in TAQ retail data. The left panel focuses on net buying, while the right panel focuses on net selling. The bars represent the time-series average of these daily shares, expressed in percentage terms. The error bars represent 95% confidence intervals computed across trading days.

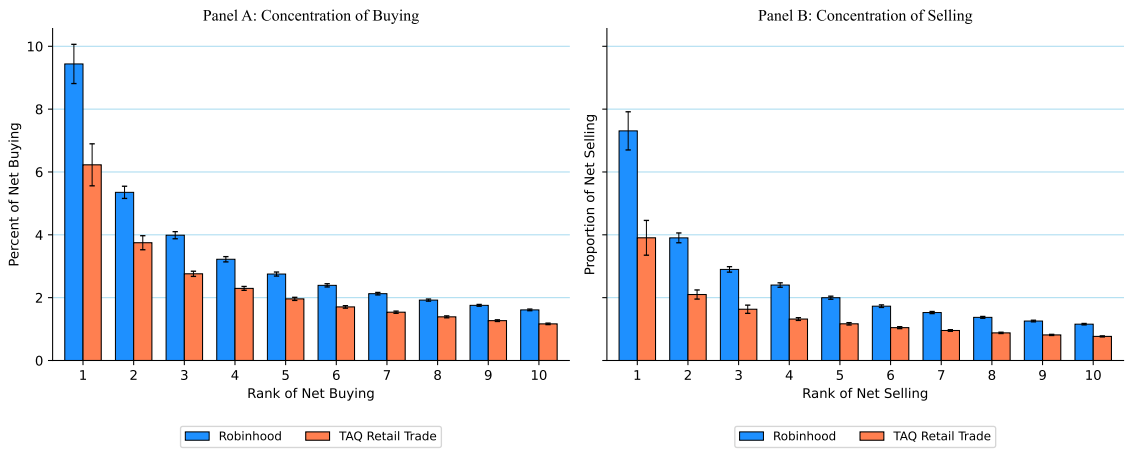


Figure 3 confirms the claims of the literature that retail trading activity is highly concentrated in a small number of stocks and more so on the buy side. On the buy side (Panel A), the most heavily traded stock accounts for 9.44% of daily net buying on Robinhood and 6.23% in TAQ retail data, while the top ten stocks together account for 34.51% and 23.93%, respectively. On the sell side, the most heavily traded stock accounts for 7.31% of daily net selling on Robinhood and 3.90% in TAQ retail data, while the top

[2] This separation is primordial to characterize how concentrated retail buying and retail selling are within each day and not to observe netting across purchases and sales.

[3] To quantify sampling uncertainty, I compute standard errors across trading days and construct 95% confidence intervals using the conventional normal approximation.

ten stocks together account for 25.55% and 14.53%, respectively. In line with Barber et al. (2022), Figure 3 also suggests that retail trading is more concentrated for Robinhood users than the broader retail investor population captured in TAQ data. This apparent material difference in concentration levels may be attributable to the facts reported in subsection 3.2.1: Robinhood users are generally younger and more inexperienced that the average retail investor[4], thus plausibly more biased towards attention-grabbing stocks (Seasholes & Wu, 2007). Overall, these findings suggest that retail investors' search for attention-grabbing stocks leads to a high degree of concentration in their trading activity, particularly on the buy side, and that this concentration is even more pronounced among Robinhood users.

Finally, to corroborate the rank-based evidence above, I compute—also following Barber et al. (2022)—two complementary daily concentration measures for retail trading activity: the Herfindahl-Hirschman Index (HHI) and the share of aggregate activity accounted for by the ten largest stock-level flows. The HHI is defined formally in Equation 4.1.

$$\mathrm{HHI}_t = \sum_{i=1}^{N_t} s_{i,t}^2 \tag{4.1}$$

In Equation 4.1, $s_{i,t}$ denotes the share of stock i in total retail net buying or total retail net selling on day t, and N_t is the number of stocks with non-zero activity on the relevant side of the market that day. A higher HHI indicates that trading is concentrated in fewer tickers. I compute this statistic separately for Robinhood user changes and TAQ retail net buying, and separately for purchases and sales. On the buy side, I retain only stocks with strictly positive net flow and scale each stock's flow by the cross-sectional total of positive flows on that day. On the sell side, I retain only stocks with strictly negative net flow, convert these flows into positive magnitudes, and again normalize by the corresponding cross-sectional total. This yields, for each day and for each dataset, a complete cross-sectional distribution of shares from which the HHI is calculated. In parallel, I compute the cumulative share of daily activity accounted for by the ten largest stocks on each side of the market. The HHI and Top-10 measures are therefore complementary: the former summarizes the concentration of the entire cross-section, whereas the latter isolates the economic importance of the largest trading destinations.

Table 4 reports the time-series means of these daily concentration statistics for Robinhood and TAQ, together with standard errors computed across trading days. To ensure that the comparison between the two datasets is not driven by differences in time coverage, I first construct the daily statistics separately for Robinhood and TAQ and then merge them by date, so that the difference between the two is evaluated on the common daily sample only. I report four objects for each dataset: the HHI of net buying, the fraction of net

[4] https://www.nytimes.com/2020/07/08/technology/robinhood-risky-trading.html.

Table 4. Concentration of Retail Buying and Selling

The sample consists of stocks for which both Robinhood user changes and TAQ retail net buying are observed. Statistics are computed from daily concentration measures and reported as time-series averages across trading days. In Panel A, *HH_buy* is the Herfindahl-Hirschman index for stocks with net buying, defined as the sum of squared shares of buying, and *Top10_buy* is the percentage of all net buying accounted for by the 10 stocks with the highest level of net buying. In Panel B, *HH_sell* is the Herfindahl-Hirschman index for stocks with net selling, defined as the sum of squared shares of selling, and *Top10_sell* is the percentage of all net selling accounted for by the 10 stocks with the highest level of net selling. Standard errors, reported in parentheses, are computed from the time-series variation in the daily statistics. The difference column reports Robinhood minus TAQ on the common daily sample. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Robinhood User Changes	TAQ Retail Trades	Difference (RH − TAQ)
Panel A: Mean Daily Concentration Measures among Stocks with Net Buying			
HH_buy (bps)	261.414***	170.726***	135.785***
	(22.105)	(32.814)	(11.326)
Top10_buy (%)	34.512***	23.933***	10.999***
	(0.423)	(0.411)	(0.356)
Panel B: Mean Daily Concentration Measures among Stocks with Net Selling			
HH_sell (bps)	173.416***	92.123***	123.833***
	(19.703)	(26.631)	(19.566)
Top10_sell (%)	25.550***	14.531***	11.492***
	(0.380)	(0.364)	(0.386)

buying attributable to the ten most heavily traded stocks, the HHI of net selling, and the corresponding Top-10 share for selling. For each statistic, I also report the daily mean difference between Robinhood and TAQ and assess its statistical significance using the time-series variation of the date-by-date differences. This procedure provides a compact summary of whether the greater concentration suggested by Figure 3 is a persistent feature of the data rather than a purely visual pattern.

From Table 4, the visual patterns in Figure 3 are confirmed more formally. On the buy side, the HHI of net buying is 261.414 bps for Robinhood and 170.726 bps for TAQ, while the Top-10 share equals 34.512% and 23.933%, respectively. On the sell side, the HHI of net selling is 173.416 bps for Robinhood and 92.123 bps for TAQ, while the Top-10 share equals 25.550% for Robinhood and 14.531% for TAQ. The Robinhood–TAQ difference is statistically significant at the 1% level for all four measures. These results confirm two central features of the data: retail trading is more concentrated on the buy side than on the sell side, and Robinhood trading is systematically more concentrated than the broader retail activity captured in TAQ. This conclusion is, of course, conditional on the sample period under study, which overlaps with the COVID-19 episode and may therefore reflect a particularly unusual composition of retail market participants. Subject to that qualification, the evidence is consistent with Robinhood users exhibiting a stronger tendency to concentrate their trading in attention-grabbing stocks, making the mechanisms studied in this thesis especially pronounced in that setting. The next sections investigate

the origins of this concentration, namely how attention is funneled toward a small set of highly salient tickers.

4.2. Determinants of User Flows

In a first attempt to understand the drivers of the concentration of retail trading activity, I examine the relationship between Robinhood daily stock-level user flows $userchg_{i,t}$ and a lagged measure of abnormal Google search attention $asvi_{i,t-1}$[5], controlling for several lagged variables within a nested linear regression framework. More formally, I estimate variations of the specification in Equation 4.2.

$$userchg_{i,t} = \alpha_i + \delta_t + \beta asvi_{i,t-1} + \theta \mathbf{X}_{i,t-1} + \varepsilon_{i,t} \tag{4.2}$$

In Equation 4.2, α_i and δ_t are stock and date fixed effects, respectively, $\mathbf{X}_{i,t-1}$ is a vector of lagged control variables, and $\varepsilon_{i,t}$ is the error term. The coefficient of interest is β, which captures the relationship between search attention and user flows after controlling for confounding factors. I estimate this specification using ordinary least squares (OLS) with with standard errors two-way clustered by stock and date to account for heteroskedasticity and dependence in the error term $\varepsilon_{i,t}$. The design purposefully exploits lagged variables to emphasize the role each regressor plays in *predicting* user flows, rather than describing contemporaneous comovements.

4.2.1. Control Variables

Before estimating the regressions described by Equation 4.2, I first describe more precisely the lagged control variables included in $\mathbf{X}_{i,t-1}$. To capture the strongly documented tendency of retail investors to chase past returns (Ardia et al., 2023; Barber & Odean, 2008; Eaton et al., 2022), I construct the binary variable $top_20_{i,t-1}$ which equals one if stock i belongs to the set of the 20 stocks with the largest absolute return on day $t-1$, and zero otherwise. I also include—aligned with the attention literature—a lagged measure of abnormal trading volume defined in a manner analogous to $asvi_{i,t-1}$, which I denote by $abvol_{i,t-1}$ (see Equation 4.3).

$$abvol_{i,t-1} = \ln\left(1 + \frac{volume_{i,t-1}}{\frac{1}{20}\sum_{\tau=2}^{21} volume_{i,t-\tau}}\right) \tag{4.3}$$

Furthermore, to account for persistence in user flows, I include the lagged dependent variable $userchg_{i,t-1}$ as a control. Finally, I also add lagged controls from RavenPack related to news coverage and sentiment. These variables have been described in section 3.4.

[5] The abnormal search attention variable $asvi_{i,t-1}$ is defined in Equation 3.1 and captures the lagged abnormal search interest for a stock i on the previous day $t-1$.

4.2.2. Estimation & Results

Table 5. Determinants of Robinhood User Flows

The table examines the determinants that predict the Robinhood $userchg_{i,t}$ variable ($users_close_t - users_close_{t-1}$) using linear regressions estimated by OLS with fixed effects for stock and date. $asvi_{i,t-1}$ is the lagged measure of abnormal search volume. $top_20_{i,t-1}$ is an indicator that equals one if the absolute return is ranked in the top 20 on day $t-1$. $abvol_{i,t-1}$ is the logarithm of the ratio of stock market volume on day $t-1$ to the average volume from days $t-21$ to $t-2$. $userchg_{i,t-1}$ is the lagged change in Robinhood users. $num_news_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$. $num_news_relevant_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$ for which RavenPack assigns a relevance score of at least 75. $ess_{i,t-1} \in [-1, 1]$ is the lagged Event Sentiment Score of RavenPack. $css_{i,t-1} \in [-1, 1]$ is the lagged and weighted by relevance Composite Sentiment Score of RavenPack. $anl_chg_{i,t}$ captures the lagged change in analyst recommendations for stock i on day $t-1$ (see section 3.4 for more details on RavenPack variables). The sample used to estimate these regressions is the attention sample, i.e. the subset of the baseline trading sample for which Google search volume data are available. Robust standard errors reported in parentheses are clustered by day and stock. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	$userchg_{i,t}$			
	(1)	(2)	(3)	(4)
$asvi_{i,t-1}$	2.406***	−0.960***	−0.951***	−0.954***
	(0.606)	(0.298)	(0.298)	(0.298)
$top_20_{i,t-1}$	60.558***	21.762***	21.782***	21.777***
	(12.027)	(6.776)	(6.778)	(6.776)
$abvol_{i,t-1}$	56.192***	−0.008	0.068	0.182
	(8.952)	(3.662)	(3.667)	(3.664)
$userchg_{i,t-1}$		0.603***	0.603***	0.603***
		(0.046)	(0.046)	(0.046)
$num_news_{i,t-1}$			0.0002*	0.0002*
			(0.0001)	(0.0001)
$num_news_relevant_{i,t-1}$			−0.0018**	−0.0019**
			(0.0009)	(0.0008)
$ess_{i,t-1}$				−9.4544***
				(2.954)
$css_{i,t-1}$				12.648**
				(6.436)
$anl_chg_{i,t-1}$				169.70
				(152.23)
Observations	934,966	934,966	934,966	934,966
R^2	0.0032	0.3636	0.3636	0.3637
Entity FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

Table 5 reports estimates of Equation 4.2 across a sequence of increasingly demanding specifications. The coefficient on $asvi_{i,t-1}$ is statistically significant throughout, but its sign reverses once the lagged dependent variable $userchg_{i,t-1}$ is introduced. This pattern suggests that abnormal search intensity is closely tied to highly persistent short-run flow dynamics. In particular, the strong positive coefficient on $userchg_{i,t-1}$ indicates substantial persistence in Robinhood user flows[6], so that the positive relation between search attention

[6] In unreported regressions that reproduce excatly the specification in column (4) but with two additional lags of $userchg_{i,t-\tau}$ for $\tau = 5, 10$, I find that the all the lags are statistically significant at the 1% level and

and subsequent inflows in the parsimonious specification appears largely to reflect this persistence. Conditional on past order-flow pressure, the remaining variation in $asvi_{i,t-1}$ is instead associated with lower next-day inflows, consistent with a reversal or timing effect rather than with sustained incremental buying pressure. An analogous pattern arises for $abvol_{i,t-1}$, which is strongly positive in the baseline specification but becomes economically negligible and statistically insignificant once lagged user flows are controlled for[7]. By contrast, the coefficient on $top_20_{i,t-1}$ remains positive and highly significant across all columns, implying that stocks with extreme lagged returns attract roughly 22 additional Robinhood users on the following day. This effect is economically meaningful and aligned with the view that extreme price moves make stocks salient to retail investors, thereby drawing incremental attention and participation (Barber & Odean, 2008; Barber et al., 2022; Eaton et al., 2022). The RavenPack sentiment variables, $css_{i,t}$ and $ess_{i,t}$, are also significant predictors of user flows but with opposite signs, thus warranting caution in their interpretation[8]. Finally, the buy and sell side regressions reported in Appendix A.2 sharpen the analysis. Persistence is almost entirely a buy-side phenomenon, with $userchg_{i,t-1}$ loading strongly in the buy specification but becoming insignificant in the sell specification, whereas salience variables such as $top_20_{i,t-1}$ and $abvol_{i,t-1}$ remain positively associated with both inflows and the magnitude of outflows, indicating that attention reallocates retail trading toward a small set of highly visible stocks. Importantly, $asvi_{i,t-1}$ positively predicts sell-side outflows, implying that high lagged search attention for a given stock is associated with more users exiting that stock the following day[9]. Contrastingly, statistical significance vanishes on the buy side. This matter will be further developed at a later stage since it probably reflects a deeper insight about the psychology of retail investors and the role of search attention in their trading decisions. Overall, these results nuance previous observations in the literature (Da et al., 2011) and reveal that lagged search attention—as measured by abnormal Google search volume—is mostly contemporaneously associated with user flows, thus suggesting that it serves as a *confirmatory* proxy of the salience of a particular stock i at time t rather than testifying of *anticipatory* buying pressure. However, the relatively small economic magnitude of the coefficients on $asvi_{i,t-1}$ in the

remain economically meaningful with coefficients of 0.508, 0.182, and 0.059 for $\tau = 1, 5, 10$, respectively.

[7] In further unreported regressions that reproduce excatly the specification in column (4) but exclude the lagged dependent variable $userchg_{i,t-1}$, I find that the coefficients on both $asvi_{i,t-1}$ and $abvol_{i,t-1}$ are positive, 2.391 and 56.369 respectively, and highly significant at the 1% level, thus confirming that the reversal pattern observed in Table 5 is indeed driven by the inclusion of lagged user flows.

[8] The time correlation between $css_{i,t}$ and $ess_{i,t}$ is 0.6766 but the panel correlation across ticker-date observations drops to 0.2143, so multicollinearity should not be a concern. However, the fact that these two sentiment measures have opposite signs and similar magnitudes suggests that they may be capturing different aspects of the news environment, or that one of them may be picking up noise rather than true sentiment. Further analysis would be needed to disentangle these effects but since the construction of these variables depends on proprietary RavenPack algorithms and do not consitute the focus of my analysis I do not investigate this issue further.

[9] See Table 12 in section A.2 for more details on the sell-side regression results.

fully specified regressions advise circumspection and call for further research on the role of search attention in retail trading activity, which is the focus of the next sections.

4.3. Determinants of Herding Events

4.3.1. Estimation & Main Results

To complement the linear user-flow regressions, I now turn to linear probability models (LPMs) specifications of herding events[10]. These specifications address a distinct question: rather than asking whether abnormal search attention predicts the average magnitude of next-day user flows, they test whether elevated search attention increases the probability that a stock enters a discrete herding state. For comparability and consistency, the design of these specifications is similar to the one of the user-flow regressions and exploits the exact same subset of the data, to the exception of the dependent variable $rh_herd_{i,t}$ which is now defined as a binary indicator of whether stock i is in a herding state on day t[11]. More formally, I estimate variations of Equation 4.4.

$$rh_herd_{i,t} = \alpha_i + \delta_t + \beta asvi_{i,t-1} + \theta \mathbf{X}_{i,t-1} + \varepsilon_{i,t} \tag{4.4}$$

Table 6 reports estimates of Equation 4.4 across a sequence of increasingly demanding specifications. The coefficient on $asvi_{i,t-1}$ is positive and statistically significant across all specifications, indicating that a one-unit increase in $asvi_{i,t-1}$ is associated with a 0.02 percentage-point increase in the probability that stock i enters a herding state on day t, holding constant stock fixed effects, date fixed effects, and the other covariates. The economic magnitude of this effect is small but non-negligible, especially considering that the average daily probability of a herding event in the sample is 0.5% by construction. These results are aligned with the ones observed in Barber et al. (2022) and subsit even after controlling for lagged herding events, or lagged user flows[12]. The direction of the effect should not be interpreted in opposition to the pattern described in Table 5, where the coefficient on $asvi_{i,t-1}$ becomes negative once lagged user flows are controlled for. The linear regressions and the LPMs address different empirical questions and the LPMs are especially focused in identifying the role of search attention in extreme relative changes in user flows, thereby capturing tail behaviors.

Turning to the other covariates, the coefficient on the lagged dependent variable $rh_herd_{i,t-1}$ follows our earlier observations and shows that a stock experiencing a herding

[10] An approach based on LPM specifications is also followed by Barber et al. (2022).

[11] For the precise definition of the herding state variable $rh_herd_{i,t}$, see section 3.5.

[12] Table 6 still demonstrates that the inclusion of $rh_herd_{i,t-1}$ diminishes the magnitude of the coefficient on $asvi_{i,t-1}$, a pattern similar to the one observed in Table 5 when $userchg_{i,t-1}$ is included. Unreported results show that in the LPMs, the inclusion of $userchg_{i,t-1}$ alone does not significantly affect the coefficient on $asvi_{i,t-1}$, whereas the inclusion of $rh_herd_{i,t-1}$ alone reduces the coefficient on $asvi_{i,t-1}$ by about 50%.

Table 6. Determinants of Robinhood Herding Events

The table examines the determinants that predict the Robinhood $rh_herd_{i,t}$ variable (constructed as in Barber et al. (2022), it is a binary indicator taking the value of 1 if a stock is in the top 0.5% of positive user change ratio on day t and a minimum of 100 users on day $t-1$) using linear probability models estimated by OLS with fixed effects for stock and date. $asvi_{i,t-1}$ is the lagged measure of abnormal search volume. $top_20_{i,t-1}$ is an indicator that equals one if the absolute return is ranked in the top 20 on day $t-1$. $abvol_{i,t-1}$ is the logarithm of the ratio of stock market volume on day $t-1$ to the average volume from days $t-21$ to $t-2$. $userchg_{i,t-1}$ is the lagged change in Robinhood users. $rh_herd_{i,t-1}$ is the lagged herding indicator. $num_news_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$. $num_news_relevant_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$ for which RavenPack assigns a relevance score of at least 75. $ess_{i,t-1} \in [-1, 1]$ is the lagged Event Sentiment Score of RavenPack. $css_{i,t-1} \in [-1, 1]$ is the lagged and weighted by relevance Composite Sentiment Score of RavenPack. $anl_chg_{i,t}$ captures the lagged change in analyst recommendations for stock i on day $t-1$ (see section 3.4 for more details on RavenPack variables). The sample used to estimate these regressions is the attention sample, i.e. the subset of the baseline trading sample for which Google search volume data are available. Robust standard errors reported in parentheses are clustered by day and stock. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	$rh_herd_{i,t}$			
	(1)	(2)	(3)	(4)
$asvi_{i,t-1}$	0.0004***	0.0002***	0.0002***	0.0002***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
$top_20_{i,t-1}$	0.0091***	0.0057***	0.0057***	0.0057***
	(0.0009)	(0.0007)	(0.0007)	(0.0007)
$abvol_{i,t-1}$	0.0115***	0.0078***	0.0078***	0.0078***
	(0.0007)	(0.0005)	(0.0005)	(0.0005)
$rh_herd_{i,t-1}$		0.123***	0.123***	0.123***
		(0.0076)	(0.0076)	(0.0076)
$userchg_{i,t-1}$		0.0000*	0.0000*	0.0000*
		(0.0000)	(0.0000)	(0.0000)
$num_news_{i,t-1}$			0.0000	0.0000
			(0.0000)	(0.0000)
$num_news_relevant_{i,t-1}$			0.0000	0.0000
			(0.0000)	(0.0000)
$ess_{i,t-1}$				0.0001
				(0.0003)
$css_{i,t-1}$				0.0023*
				(0.0013)
$anl_chg_{i,t-1}$				-0.0344
				(0.0418)
Observations	934,966	934,966	934,966	934,966
R^2	0.0061	0.0208	0.0208	0.0208
Entity FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

event on day $t-1$ is 12.3 percentage points more likely to experience another herding event on day t, thus confirming the strong persistence in herding events documented in Barber et al. (2022). The rationale for this persistence could be explained by the fact that the herding event generates social media attention, which in turns propagate into heightened likelihood of another herding event (De Giorgi et al., 2026). The coefficient on $top_20_{i,t-1}$ is also positive and significant, indicating that stocks with extreme lagged returns are more likely to experience buying pressures. The coefficient on $abvol_{i,t-1}$ also points out that stocks with elevated lagged trading volume are more likely to experience herding events. The remaining covariates are either not statistically significant or economically meaningful predictors of herding events in the fully specified model, except for $css_{i,t-1}$ which is positive and significant at the 10% level, thus suggesting that more positive news sentiment may be associated with a higher likelihood of herding events. This particular evidence resonates with the empirics in Chan and Fong (2004), who document that individual investors become more inclined to buy when they perceive the crowd to be optimistic. However, it diverges with Münster et al. (2024), who note that retail investors oftentimes act as contrarians by selling stocks with positive news sentiment.

4.3.2. Heterogeneity of the Attention Effect

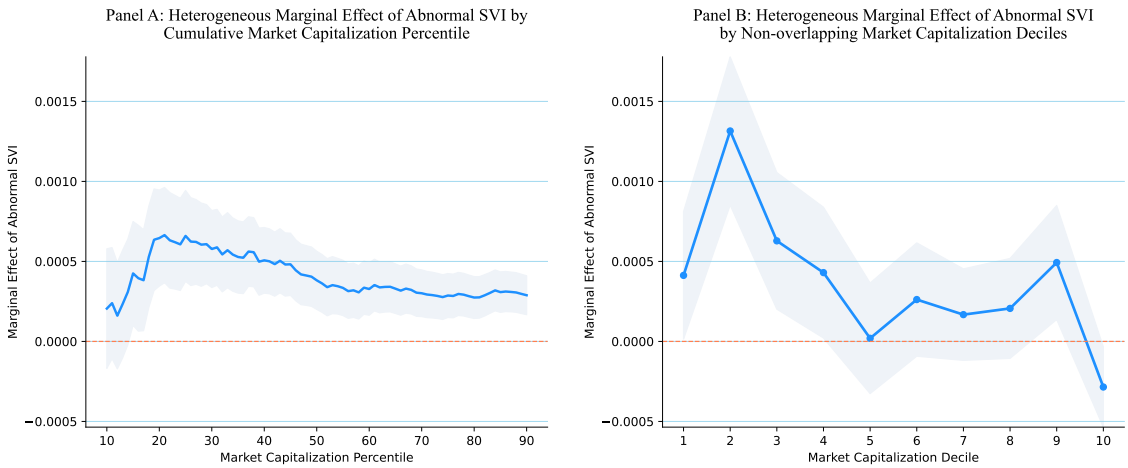
The attention literature has largely reported that the effect of attention on retail trading activity is heterogeneous across stocks with different market capitalizations, with contrasted views about the strength or direction of the effect across attention proxies (Ardia et al., 2023; Da et al., 2011; De Giorgi et al., 2026; Fedyk, 2021; Warkulat & Pelster, 2024). To contribute to this debate, I investigate whether the effect of $asvi_{i,t-1}$ on the probability of herding events displays such behaviors. To do so, I estimate variations of Equation 4.4 where I interact $asvi_{i,t-1}$ with a binary variable $size_under(k)_{i,t-1}$ that equals 1 if the stock's lagged market capitalization is under the k-th percentile and 0 otherwise. More formally, I estimate the specification in Equation 4.5 for $k \in \{10, 11, \ldots, 89, 90\}$.

$$\begin{aligned} rh_herd_{i,t} &= \alpha_i + \delta_t + \beta_1 asvi_{i,t-1} \\ &+ \beta_2 asvi_{i,t-1} \times size_under(k)_{i,t-1} + \theta \mathbf{X}_{i,t-1} + \varepsilon_{i,t} \end{aligned} \tag{4.5}$$

It should be noted that the presence of date fixed effects δ_t in Equation 4.5 implies that adding the main effect $size_under(k)_{i,t-1}$ would lead to perfect multicollinearity whenever the binary variable $size_under(k)_{i,t-1}$ is time-invariant for a given stock i (i.e. if the stock's market capitalization is always above or always below the k-th percentile during the sample period). Therefore, to avoid this issue and ensure comparability with previous specifications, I do not include $size_under(k)_{i,t-1}$ in the regressions and instead focus on the partial effect of $asvi_{i,t-1}$ for stocks with market capitalization under the k-th percentile, which is given by $\beta_1 + \beta_2$. I compute the corresponding standard error using the delta

method, taking into account both coefficient variances and their covariance, and construct 90% confidence intervals based on the asymptotic Gaussian approximation. Panel A of Figure 4 plots this construction.

Figure 4. Heterogeneity of the Attention Effect. Panel A plots the partial effect of $asvi_{i,t-1}$ on the probability of herding events for stocks with market capitalization under the k-th percentile. The x-axis represents the percentile threshold k and the y-axis represents the partial effect of $asvi_{i,t-1}$ for stocks with market capitalization under the k-th percentile. The shaded area represents the 90% confidence interval for the partial effect. Standard errors are computed by clustering by stock and date and using the delta method to account for the covariance between β_1 and β_2. Panel B plots the partial effect of $asvi_{i,t-1}$ on the probability of herding events for each market-cap decile. The construction of the confidence intervals is analogous to the one in Panel A.



Panel A shows that when small-cap stocks are defined cumulatively, the marginal effect of $asvi_{i,t-1}$ on next-day Robinhood herding is positive throughout and is strongest in the lower tail of the size distribution, peaking around the 20th to 25th market-cap percentile before gradually declining as the threshold expands to include larger firms. This pattern suggests that attention-induced herding is concentrated among relatively small stocks and becomes diluted when progressively larger firms are added to the treated group.

To confirm this interpretation, I estimate another LPM characterized by Equation 4.6.

$$\begin{aligned} rh_herd_{i,t} &= \alpha_i + \delta_t + \beta\, asvi_{i,t-1} \\ &+ \sum_{d \neq 10} \gamma_d \big(asvi_{i,t-1} \times 1\{decile = d\} \big) + \theta X_{i,t-1} + \varepsilon_{i,t} \end{aligned} \tag{4.6}$$

Market capitalization is partitioned into ten mutually exclusive deciles using the full cross-sectional distribution of *size*, where decile 1 contains the smallest-cap observations and decile 10 the largest-cap observations. To allow the effect of abnormal search attention to vary flexibly across size groups, I interact $asvi_{i,t-1}$ with indicator variables for each market-cap decile, omitting the largest-cap decile as the reference category. Standard errors are two-way clustered by stock and date. Because the coefficient on $asvi_{i,t-1}$ captures the effect for the omitted reference group, namely the largest-cap decile, the marginal effect for

any other decile is recovered as the sum of the coefficient on $asvi_{i,t-1}$ and the coefficient on the corresponding interaction term. I compute confidence interval with the same method as in Panel A. This procedure yields a decile-specific profile of the sensitivity of Robinhood herding probabilities to lagged abnormal search attention across the market-cap distribution. Panel B finally confirms our interpretation of Panel A using non-overlapping market-cap deciles. The effect is largest in the second decile and remains positive for several lower and middle deciles, whereas it is close to zero or negative in the largest-cap segment. Taken together, the two panels indicate that the attention-to-herding channel is not uniform across stocks, but is instead most pronounced among smaller caps. A possible explanation is that attention-induced herding should be strongest where retail demand has the greatest price impact and where arbitrage is least able to offset promptly the resulting mispricing (Garcia, 2026). Smaller-cap stocks tend to be less liquid (Amihud, 2002), less institutionally followed (Merton, 1987)[13], and more prone to display lottery-like characteristics, thus attracting more retail investors (Kumar, 2009); so a given attention spike is more likely to translate into concentrated retail order flow and a measurable herding episode.

[13] In Merton (1987) model, investors do not all know or follow the same set of securities. Some stocks are "neglected", in the sense that they are held in the information or consideration sets of fewer investors. Because fewer investors pay attention to them, those stocks are held less widely, their idiosyncratic risk is thus shared by a smaller pool of investors, and investors could require additional expected return as compensation for bearing that risk. The link to my attention-induced herding analysis is that a stock's investor base and visibility are themselves economically important, hence, spikes shocks to visibility and attention should have stronger effects on their demand and pricing.

v. Timing of the Attention Process

The present chapter turns to the timing of Google search attention itself. The evidence produced in chapter 4 thus far indicates that abnormal search intensity is linked to Robinhood trading activity, yet its role appears more nuanced than that of a simple *leading* indicator of retail demand. In the linear user-flow regressions, the predictive content of lagged $asvi_{i,t-1}$ is largely absorbed by the persistence of flow dynamics and becomes negative once these are controlled for—both for Robinhood and TAQ retail measures, whereas in the linear probability models it remains positively associated with the incidence of discrete herding events—as in Barber et al. (2022), especially among smaller-cap stocks. This pattern suggests that search attention may not primarily precede retail trading, but may instead rise contemporaneously with the salience shocks that place a stock in investors' consideration sets and ultimately culminate in herding. Against this background, this chapter conducts a brief event-study analysis of Google search attention from two different angles in order to characterize more precisely whether it behaves as an *anticipatory* signal of retail trading or rather as a *confirmatory* manifestation of heightened stock salience. Importantly, this chapter also serves as prefatory work for the next one, which will focus on causal links and shed light on the central claim of this thesis.

5.1. Event-Study around Search Attention Spikes

First, I examine the dynamic behavior of Robinhood user flows around unusually large attention shocks[1] by designing an event-study in line with some guidelines of MacKinlay (1997). Attention is measured across stock i and day t as before with $asvi_{i,t}$. To avoid a mechanical contemporaneous overlap between the event definition and the flow response, events are defined using lagged attention. Specifically, for each stock i, I compute the empirical 99th percentile of the historical distribution of $asvi_{i,t-1}$, $\hat{q}_{i,0.99}$, and classify a stock-date as a high-attention event candidate whenever $asvi_{i,t-1} > \hat{q}_{i,0.99}$. This stock-level definition is favored over a pooled cutoff to allow for persistent heterogeneity in the unconditional level and volatility of search activity across tickers and ensures that an event corresponds to an unusually large within-stock attention shock. Because elevated attention can persist for several consecutive trading days, I define the event date as the onset of such an episode, i.e. the first day in a run of stock-dates satisfying the event-candidate

[1] Importantly, the word "shock" is used here in a purely descriptive sense and interpreted as a synonym of "spike", "surge", or "burst". Therefore, an attention shock is not necessarily understood as an exogenous surprise, but rather as a salient episode of elevated attention that may be driven by endogenous or exogenous factors. The event-study design is purely descriptive and does not rely on any exogeneity assumption about the timing of attention shocks.

criterion. I then impose a strict non-overlap restriction: once an event is retained for a given stock, no subsequent event for that stock may occur within the next $2W + 1$ trading days, where $W = 10$ denotes the half-window length. This restriction guarantees that the event windows are not contaminated by adjacent attention episodes and allows each retained event to be interpreted as a distinct salience shock. Critically, antecedent empirical evidence in chapter 4 suggests that herding events are marked by a significant persistence, thus plausibly cluster in time. The non-overlap restriction is therefore a guarantee that the event-study design captures the path of retail trading around a single search attention spike rather than a sequence of closely spaced episodes, but is also ultimately a limitation in the sense that the possible clustering behavior of herding events is not examined. Around each retained event, I construct a balanced event-time window over trading days $\tau \in [-10, +10]$. Only complete windows are kept in the final sample, so that changes in the estimated event-time path are not driven by variation in sample composition across horizons. This procedure yiels a final sample of 7,542 unique events.

Subsequently, to isolate abnormal retail trading from persistent stock-specific and marketwide components, the Robinhood user-flow series is residualized with respect to stock and date fixed effects, as depicted in Equation 5.1.

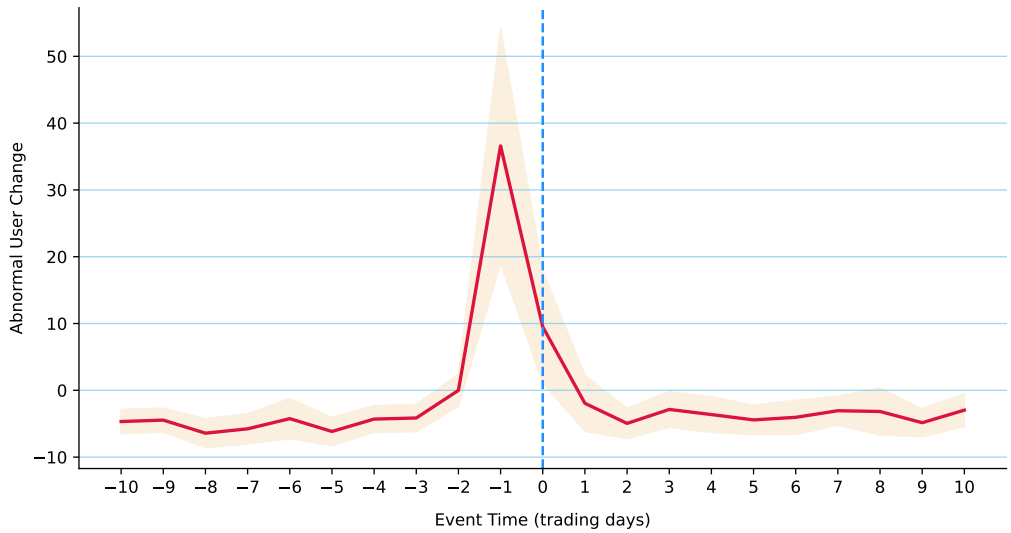
$$userchg_{i,t} = \alpha_i + \delta_t + \varepsilon_{i,t} \tag{5.1}$$

As usual, $userchg_{i,t}$ denotes the daily change in the number of Robinhood users holding stock i, α_i are stock fixed effects, and δ_t are date fixed effects. The residual $\varepsilon_{i,t}$ is interpreted as abnormal user flow, purged of time-invariant cross-sectional differences in stock popularity and of aggregate shocks common to all stocks on a given day. The event study is then conducted on these residualized flows rather than on raw user changes. The descriptive estimates reported in this section are event-time averages of the abnormal user-flow residuals. For each relative event time $\tau \in \{-10, \ldots, 10\}$, I compute the mean of $\varepsilon_{i,\tau}$, $\overline{\varepsilon}_\tau$, across all retained events, together with the corresponding cross-event standard error and confidence interval. This procedure yields a nonparametric summary of the typical path of abnormal Robinhood participation before and after a large attention shock. Because the event indicator is based on $asvi_{i,t-1}$, the resulting profile is informative about whether retail trading tends to rise only after attention has already become unusually high or whether abnormal flows are already present in the days preceding the attention episode[2].

Figure 5 reveals a sharp and highly localized pattern in Robinhood user flows around large search-attention bursts. In the days preceding the event, abnormal user flows are systematically negative. From $\tau = -10$ to $\tau = -3$, $\overline{\varepsilon}_\tau$ remains in a relatively narrow range

[2] The design is therefore purely descriptive, but provides a method to characterize the timing of retail participation around salient bursts of Google search attention and serves as an intuitive bridge to the causal analysis developed in the subsequent chapter.

Figure 5. Event-Time Path of Abnormal Robinhood User Flows around Search Attention Spikes. This figure depicts the average path of abnormal Robinhood user flows around large search attention bursts (the top 1% of $asvi_{i,t-1}$, specific to each stock). The event date ($\tau = 0$) is defined as the first day in a run of stock-dates where lagged abnormal search volume $asvi_{i,t-1}$ exceeds the empirical 99th percentile of its historical distribution for stock i. The sample includes 7,542 events. Abnormal user flows are obtained as residuals from a regression of daily user changes on stock and date fixed effects. The solid line represents the average abnormal flow at each relative event time τ, while the shaded area corresponds to the 90% confidence interval based on cross-event standard errors.



between roughly −4 and −6 users per stock-day and is statistically distinguishable from zero throughout. At $\tau = -2$, the estimate becomes essentially zero and loses statistical significance. This pre-event pattern indicates that stocks experiencing an extreme spike in Google search attention are not preceded by abnormal Robinhood buying pressure over a long horizon. If anything, they are characterized by below-average retail demand up to shortly before the event. In that sense, the figure provides little support for the view that elevated search attention gradually anticipates a sustained build-up in Robinhood participation.

The most notable feature of Figure 5 is the sharp discontinuous jump between $\tau = -2$ and $\tau = -1$. On the day immediately preceding the event, abnormal Robinhood user flows rise to 36.6 users, with a 90% confidence interval of $[19.05, 54.14]$, by far the largest estimate in the event window. This surge is economically and statistically large relative to the remainder of the path and suggests that the onset of an attention burst is preceded by an abrupt and highly concentrated increase in retail participation rather than by a smooth pre-trend. On the event day itself, $\tau = 0$, abnormal user flows remain positive at 9.5 users, with a 90% confidence interval of $[1.1, 17.9]$, but the magnitude is already much smaller than on the previous trading day. Thus, the strongest flow response does not occur after the measured attention shock has materialized. This temporal ordering is economically informative. Because the event is defined using lagged abnormal search intensity, the estimate at $\tau = 0$ corresponds to trading on the day following an extreme

realization of $asvi_{i,t-1}$. The fact that the largest abnormal inflow occurs at $\tau = -1$ therefore implies that Robinhood participation spikes, as hypothesized, *contemporaneously* with the search-attention burst itself, while the residual effect on the following day is positive but substantially attenuated.

Post-event dynamics should be interpreted with some care. Because the event-study design imposes a non-overlap restriction, the estimated path after $\tau = 0$ is conditional on the event being an isolated attention-episode onset, in the sense that no additional retained onset for the same stock may occur within the next $2W + 1$ trading days. The post-event estimates therefore do not describe the unconditional dynamics following all search-attention bursts, especially in environments where such bursts could cluster in time. Rather, they characterize the evolution of abnormal Robinhood user flows after a single retained onset that is sufficiently separated from other candidate episodes. After the modest positive estimate at $\tau = 0$, abnormal user flows quickly revert and become negative again. Beginning at $\tau = 2$, the estimates are once more significantly below zero for most horizons, generally fluctuating between -3 and -5 users per stock-day. Hence, the immediate increase in Robinhood participation around the attention spike is not followed by continued abnormal inflows. This pattern is suggestive of a short-lived burst in retail demand that rapidly dissipates and is subsequently reversed. In substance, the figure points to a particular dynamic. Large search-attention shocks are associated with a sudden and concentrated increase in Robinhood participation at the onset of the episode, followed by only a modest residual continuation on the next day and a fairly rapid reversal thereafter. The absence of a prolonged positive run-up before the event, combined with the swift decay in $\overline{\varepsilon}_\tau$ afterward, argues against interpreting Google search intensity as a slow-moving anticipatory signal of retail demand. Rather, the evidence indicates that search attention and Robinhood trading respond jointly to the same salience shock, with the strongest retail response occurring essentially simultaneously with the attention burst itself. The next event study further substantiates this claim.

5.2. Event-Study around Extreme Returns

Second, I examine the timing of abnormal Google search attention around large overnight price movements. The purpose of this complementary event study is to reverse the perspective of the previous design. Rather than asking whether Robinhood user flows rise around large attention bursts, the present exercise asks whether search attention itself spikes when a stock experiences an extreme return realization. In this way, the analysis speaks directly to whether Google search intensity behaves as an anticipatory signal of retail demand or instead as a contemporaneous manifestation of stock salience induced by sharp price movements.

To align the event definition with the institutional features of the Robinhood "Top

Movers List" mechanism[3], events are defined using the cross-sectional ranking of absolute overnight returns within each trading day. Additionally, a stock is deemed eligible on day t if $size_{i,t} \geq c$, where the baseline cutoff is $c = 300$ million—again for conforming with the "Top Movers List" feature. Among eligible stocks, I then rank $|overnight_ret_{i,t}|$ in descending order within each date and define an extreme-return event $E_{i,t}$ as in Equation 5.2.

$$E_{i,t} = 1\left\{rank_t\left(|overnight_ret_{i,t}|\right) \leq 20 \text{ and } size_{i,t} \geq c\right\}. \tag{5.2}$$

Thus, an event corresponds to a stock that belongs to the daily top 20 in absolute overnight returns within the market-cap-eligible universe. I additionally partition the event sample into positive and negative overnight return events according to the sign of $overnight_ret_{i,t}$.

As in the previous event study, I impose a non-overlap restriction in order to isolate distinct episodes. Let $W = 10$ denote the half-window length. For each stock, I retain an event at date t only if no earlier retained event for the same stock occurred within the preceding $2W$ trading days. This ensures that the final event windows, defined over $\tau \in [-10, +10]$, do not contain two retained event dates for the same stock. The restriction is useful because extreme return realizations can cluster for individual stocks during volatile periods; without such a filter, the event-time profile would be difficult to interpret as the dynamics around a single salient price shock. At the same time, as in the previous event study, the resulting post-event path is conditional on the event being sufficiently isolated from other candidate episodes. This procedure yields a 6,247 unique extreme-return events, of which 3,270 are positive and 2,977 are negative.

The outcome of interest is contemporaneous abnormal search attention, $asvi_{i,t}$, i.e. at the same calendar day as the event. Around each retained event, I then build a stacked event-time panel by collecting all stock-day observations in the interval $[t - 10, t + 10]$ and indexing them by relative event time τ. Only complete windows are retained, so that every event contributes the full set of $2W + 1 = 21$ relative trading days to the final sample.

The first set of estimates is purely nonparametric. For each relative event time τ, I compute the cross-event average of $asvi_{i,t+\tau}$ together with its cross-event standard error. This yields the average event-time path of abnormal search attention around extreme overnight return events. The corresponding confidence intervals are based on the conventional Gaussian approximation. This descriptive profile is reported in Figure 6 Panel A.

To complement these raw event-time averages, I also estimate an event-time regression

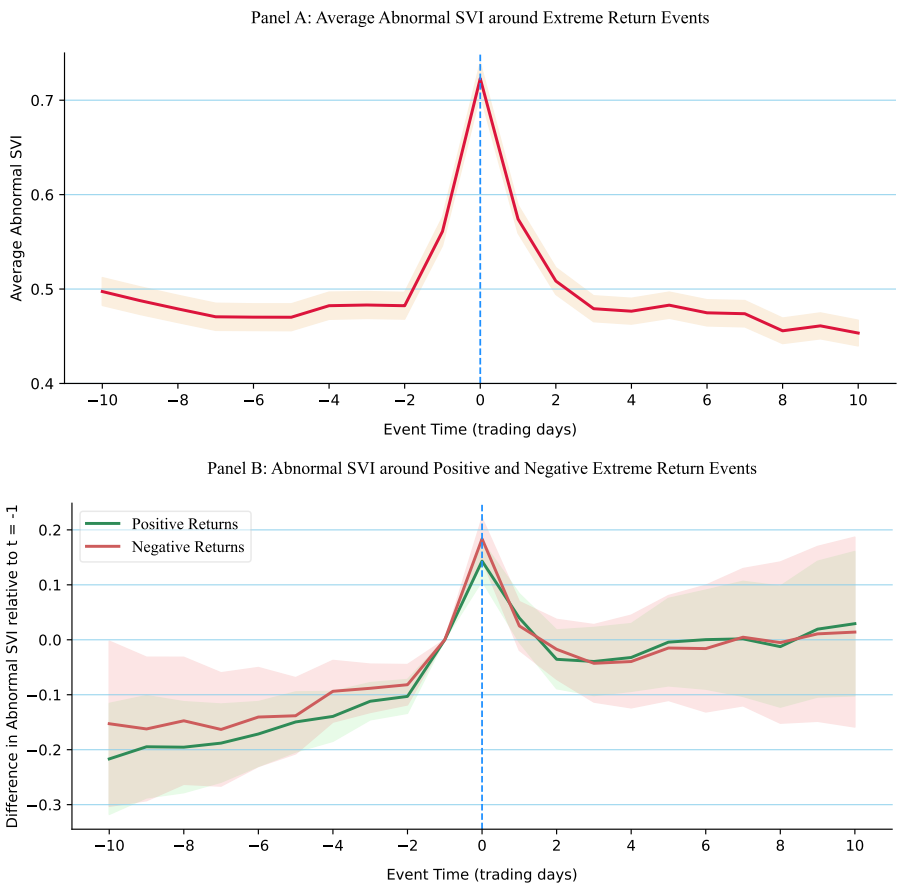
[3] The "Top Movers List" is a feature on the Robinhood platform that highlights the top 20 stocks with the highest *absolute* overnight returns and that meet the minimum market capitalization threshold of $c = 300$ million. This feature of the Robinhood app will be developed further in the next chapter.

in the stacked panel (see Equation 5.3).

$$asvi_{e,\tau} = \alpha_e + \delta_t + \sum_{\substack{k=-10 \\ k\neq -1}}^{10} \beta_k \, 1\{\tau = k\} + u_{e,\tau}, \tag{5.3}$$

Where e indexes retained events, α_e are event fixed effects, and δ_t are calendar-date fixed effects. The omitted category is $\tau = -1$, so each coefficient β_k measures the difference in abnormal search attention at relative event time k compared with the day immediately preceding the event. Event fixed effects absorb all time-invariant heterogeneity across retained events, while date fixed effects remove aggregate fluctuations in search intensity common to all stocks on a given calendar day. Standard errors are two-way clustered by event and calendar date. The outcomes of this regression are reported in Table 7 and Panel B of Figure 6 [4].

Figure 6. Event-Time Path of Abnormal Robinhood User Flows around Search Attention Spikes. Panel A depicts the average path of abnormal Google search attention around extreme overnight return events, defined as stock-dates where the absolute overnight return ranks in the top 20 among stocks with market capitalization at the open exceeding $300 million. Panel B reports the corresponding event-time regression estimates from Equation 5.3. Both panels report 90% confidence intervals.



[4] Due to display constraints, only the coefficients for $\tau = -5$ to $\tau = 5$ are reported in Table 7, but the full set of estimates is available in Appendix A.3.

Table 7. Abnormal Search Attention around Extreme Overnight Return Events

The table reports event-time regressions of abnormal Google search intensity around extreme overnight return events. An event is defined on day t when stock i belongs to the top 20 of the cross-sectional ranking of absolute overnight returns, $|overnight_ret_{i,t}|$, among stocks whose market capitalization at the open exceeds \$300 million. The dependent variable is $asvi_{i,t}$, the contemporaneous abnormal search volume measure aligned with the event date. Relative event-time indicators $1\{\tau = k\}$ are included for $k \in \{-10, \ldots, 10\} \setminus \{-1\}$, so that the omitted category is $\tau = -1$. Accordingly, each coefficient measures the difference in abnormal search attention at relative event time k compared with the day immediately preceding the event. For brevity, only the coefficients of the relative event-time from $\tau = -5$ to $\tau = 5$ are reported. All specifications include event fixed effects and calendar-date fixed effects. Robust standard errors reported in parentheses are two-way clustered by event and calendar date. Column (1) reports the full event sample. Columns (2) and (3) split the sample according to whether the overnight return is positive or negative. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	$asvi_{i,t}$		
	(1) Overall	(2) Positive Returns	(3) Negative Returns
$\tau = -5$	-0.1559^{***}	-0.1495^{***}	-0.1382^{***}
	(0.0198)	(0.0279)	(0.0353)
$\tau = -4$	-0.1273^{***}	-0.1394^{***}	-0.0937^{***}
	(0.0163)	(0.0231)	(0.0286)
$\tau = -3$	-0.1098^{***}	-0.1119^{***}	-0.0883^{***}
	(0.0128)	(0.0172)	(0.0224)
$\tau = -2$	-0.0939^{***}	-0.1030^{***}	-0.0818^{***}
	(0.0122)	(0.0157)	(0.0187)
$\tau = 0$	0.1788^{***}	0.1432^{***}	0.1837^{***}
	(0.0144)	(0.0184)	(0.0193)
$\tau = 1$	0.0463^{***}	0.0397^{*}	0.0252
	(0.0171)	(0.0225)	(0.0226)
$\tau = 2$	−0.0027	−0.0355	−0.0174
	(0.0192)	(0.0272)	(0.0278)
$\tau = 3$	−0.0157	−0.0395	−0.0429
	(0.0239)	(0.0315)	(0.0360)
$\tau = 4$	−0.0018	−0.0322	−0.0396
	(0.0236)	(0.0314)	(0.0430)
$\tau = 5$	0.0205	−0.0043	−0.0150
	(0.0278)	(0.0406)	(0.0486)
Observations	131,187	78,603	72,849
R^2	0.0781	0.0843	0.0837
Event FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Reference Date	$\tau = -1$	$\tau = -1$	$\tau = -1$

Figure 6 and Table 7 seem to deliver us a coherent narrative. Abnormal Google search attention does not display a long anticipatory build-up before extreme overnight return realizations. In the raw event-time averages reported in Panel A, average $asvi_{i,t}$ remains remarkably stable over the pre-event window, fluctuating narrowly around 0.18 to 0.20 from $\tau = -10$ to $\tau = -2$. The event-time regression sharpens this point. Relative to the day immediately preceding the event, all coefficients from $\tau = -10$ to $\tau = -2$ are negative and highly statistically significant. Because the omitted category is $\tau = -1$, these estimates imply that abnormal search attention is already elevated on the day just before the extreme return event and that this increase occurs only very late in the run-up. Put differently, the data do not support the view that Google search attention rises gradually over many days in advance of extreme price movements. Rather, attention begins to rise only shortly before the event and does so in a concentrated manner. However, due to the daily frequency of the data, it is still harduous to derive strong and definitive conclusions about the precise temporal order and caution is required in the interpretation. It is also possible that a fraction of these events are related to pre-earning announcements, which could slightly elevate search attention a couple of days before the event—i.e., just before the "surprise" component of the news generate extreme returns. This hypothesis is nevertheless weakened by the absence of any apparent gradual build-up in search attention.

The central result is the sharp contemporaneous spike at the event date. In the pooled specification, the coefficient at $\tau = 0$ is 0.1788 with a standard error of 0.0144, implying a highly statistically significant increase in abnormal search attention relative to $\tau = -1$. The same pattern is visible in Panel A, where average abnormal Google attention jumps from 0.178 at $\tau = -1$ to 3.017 at $\tau = 0$, before falling back immediately thereafter. This sharp discontinuity indicates that extreme overnight returns are accompanied by a pronounced contemporaneous burst in search activity. Economically, this is the key result of the section: abnormal Google search intensity appears, once again, to behave primarily as a *confirmation* of stock salience at the time of the return shock rather than as a slow-moving signal that precedes it by a wide margin.

The post-event profile further reinforces this interpretation. In the regression, the coefficient at $\tau = 1$ remains positive in the full sample, but is already much smaller, at 0.0463, than the event-day estimate, and from $\tau = 2$ onward the coefficients are small and statistically indistinguishable from zero. The nonparametric averages convey the same message. Although abnormal search attention remains above its pre-event level immediately after the event, it declines sharply from 3.017 at $\tau = 0$ to 0.620 at $\tau = 1$ and then continues to normalize toward its baseline range.

Panel B and Columns (2) and (3) of Table 7 show that this dynamic is present for both positive and negative overnight return events. In each case, pre-event coefficients are negative and statistically significant, and the event-day coefficient is strongly positive. The

effect is somewhat larger for negative overnight returns: the coefficient at $\tau = 0$ equals 0.1837, for negative events, compared with 0.1432 for positive events. This difference suggests that sharp overnight losses attract at least as much, and likely somewhat more, abnormal search attention than comparably extreme gains. This result is aligned with Ardia et al. (2023) who document faster reaction times from the users for negative extreme returns.

Taken together, the evidence from this second event study convincingly complements the results of the first. The previous analysis showed that Robinhood user flows spike essentially contemporaneously with large search-attention bursts. The present analysis shows, conversely, that abnormal search attention itself spikes simultaneously with extreme overnight return events. The two findings jointly point to the same conclusion: Google search attention is best understood not as a distant forward leading indicator of retail trading, but as a *confirmatory* manifestation of the salience shocks that simultaneously draw investor attention and induce retail participation. In this sense, attention appears to be part of the same immediate response process as herding, rather than a separate signal that gradually forecasts it well in advance. This claim will be further supported by the causal evidence presented in the next chapter.

VI. Manufacturing Attention

To formulate a trading decision, retail investors process a variety of signals. However, since their attention is limited and processing a signal is cognitively costly (Kahneman, 1973), they are likely to consult only a subset of the available information. The *complexity* of the signals might well determine which ones are attended to and which ones are ignored. Umar (2022) documents, unsurprisingly, that retail investors typically disregard complex information and Oprea (2020) confirms in a laboratory experiment that people display aversion to complexity. Hence, one could expect that retail investors herding will oftentimes occur in response to simple and easily digestible signals, such as an appearance on Robinhood's "Top Movers List" (Barber et al., 2022). Robinhood's Top Movers List is a continuously updated list of 20 stocks with a "market cap of more than $300 million" and displaying the "largest price movements as measured from the previous market close price"[1]. Importantly, the list does not restrict itself to positive price movements, but also includes stocks with large negative price movements, i.e. the ranking is based on absolute overnight returns. This chapter therefore exploits the structure of the Top Movers List—a signal likely to attract the attention of a broad pool of retail investors—to first replicate the central empirical findings of Barber et al. (2022) and then to sharpen the analysis of the Google search attention transmission mechanism.

6.1. Platform Salience & Stock Selection

This first section is dedicated to replicating the main empirical findings of Barber et al. (2022) regarding the impact of the Top Movers List on retail trading to establish the credibility of the signal as a driver of retail attention and trading.

6.1.1. Ranking on Absolute Price Movements

The identifying feature of Robinhood's "Top Movers" list is that it ranks stocks by the magnitude of their price movement rather than by the sign, thereby displaying extreme winners and extreme losers in a single menu. Barber et al. (2022) argue that this design choice generates a sharp comparative prediction: conditional on being a top mover, Robinhood investors should exhibit broadly similar buying responses to top gainers and top losers because both groups are made similarly *available* through the same interface. By contrast, the broader population of retail investors—whose attention is shaped by a more conventional media environment that separates winners from losers—should react more strongly to top gainers than to top losers.

[1] https://cdn.robinhood.com/disclosures/WebDisclosures.pdf.

To replicate this mechanism, I follow Barber et al. (2022) and construct, for each trading day t, the set of the 20 eligible stocks whose *absolute* overnight return from the close of day $t-1$ to the open of day t is largest, restricting to stocks whose market capitalization at the open exceeds \$300 million in line with Robinhood's stated eligibility threshold. Within this daily top-20 set, I assign a "top-mover score" $Score_{i,t} \in \{1, \ldots, 20\}$ that increases with extremity: the stock with the largest absolute move receives 20 and the 20th receives 1. I then measure buying activity on day t using an intraday Robinhood flow proxy (intraday user change)[2] and an intraday TAQ retail flow proxy (retail net buying). For expositional symmetry, I split the top-20 set into top gainers and top losers using a binary indicator $\mathbf{1}\{r_{i,t} < 0\}$.

Figure 7. Averages of Robinhood Intraday User Change Against TAQ Intraday Net Retail Buyings for Top Movers. This figure reproduces the original design of Barber et al. (2022). The panels characterize the mean Robinhood intraday user change and TAQ intraday net retail buying of day t against top mover rankings based on different measures. The rankings are sorted for stocks with market cap above \$300 million at the market open of day t. Panel A sorts top movers on absolute overnight returns of day t; Panel B sorts top movers on absolute daily returns of day t. Robinhood intraday user change measures the change in Robinhood users from first time stamp that excludes market open trades to the last time stamp before the market closes. TAQ intraday net retail buying is TAQ retail buying minus TAQ retail selling during market trading hours. The error bar represents the 90% confidence interval for the mean.

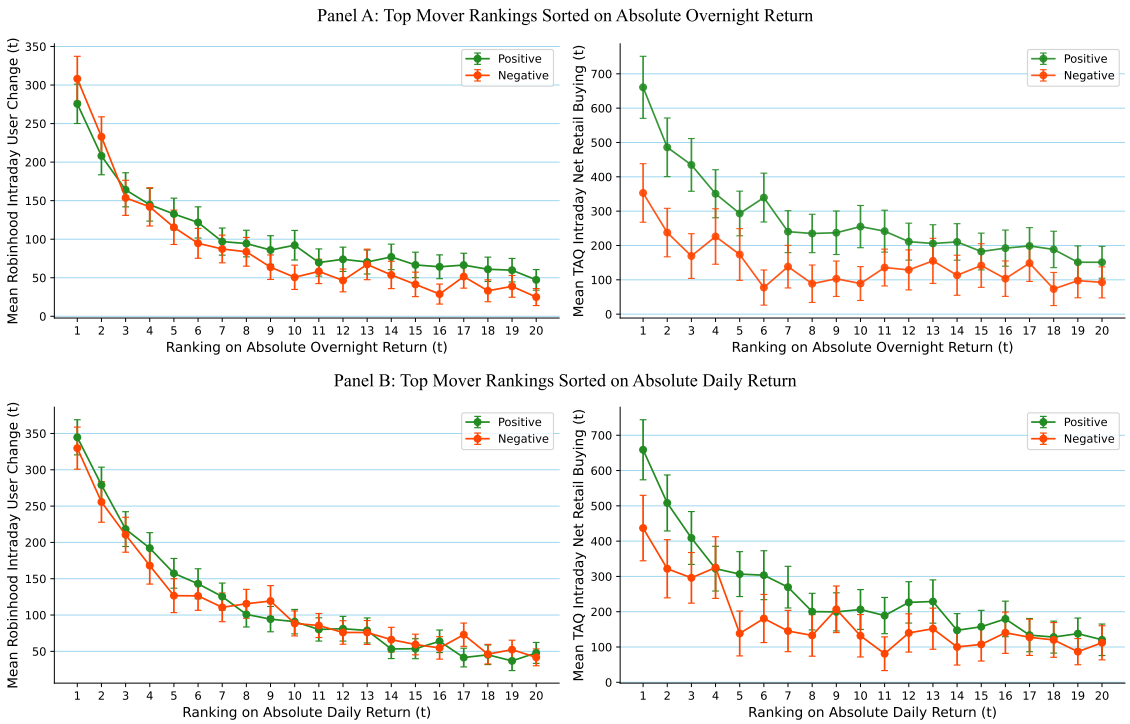


Figure 7 presents the resulting rank-response profiles. Two patterns stand out. First, buying intensity increases sharply with rank extremity in both datasets, consistent with

[2] The intraday user change is the difference between the number of Robinhood users holding stock i at the first observation after 10 a.m. ET—an approach selected to exclude market open movements, as did Barber et al. (2022)—and the number of users holding stock i at the last time stamp before the market closes.

attention-based buying in the spirit of Barber and Odean (2008). Second, the gainers–losers contrast differs markedly across platforms. Among Robinhood users, the buying response to top gainers and top losers is comparatively similar throughout the rank distribution, whereas in TAQ data retail net buying is systematically higher for top gainers than for top losers, especially among the highest-ranked movers.

To quantify these differences, I estimate the rank-based regression used by Barber et al. (2022) and expressed in Equation 6.1.

$$y_{i,t} = \delta_t + \beta_1 \, Score_{i,t} + \beta_2 \, \mathbf{1}\{r_{i,t} < 0\} + \beta_3 \, Score_{i,t} \times \mathbf{1}\{r_{i,t} < 0\} + \varepsilon_{i,t} \tag{6.1}$$

Where $y_{i,t}$ is alternatively Robinhood or TAQ buying activity for stock i on day t, the sample includes only the daily top 20 movers, δ_t are day fixed effects, and standard errors are clustered by day.

Table 8. Top Movers Rank Regressions

This table estimates Equation 6.1 on the daily top-20 mover set. $Score_{i,t}$ assigns 20 to the stock with the largest absolute return and 1 to the 20th. $\mathbf{1}\{r_{i,t} < 0\}$ is an indicator for negative returns (top losers). The dependent variable is Robinhood intraday user change (RH) or TAQ intraday net retail buying (TAQ). Panel A ranks top movers by absolute overnight returns; Panel B ranks by absolute daily returns. All specifications include day fixed effects. Standard errors (in parentheses) are clustered by day. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Robinhood User Changes	TAQ Retail Trades
	Panel A: Absolute Overnight Return Rank	
$Score_{i,t}$	8.1266^{***}	17.3109^{***}
	(0.5181)	(1.8012)
$\mathbf{1}\{r_{i,t} < 0\}$	-16.1050^{**}	36.7698
	(7.8472)	(25.1131)
$Score_{i,t} \times \mathbf{1}\{r_{i,t} < 0\}$	2.1649^{***}	-9.0662^{***}
	(0.7390)	(2.3264)
	Panel B: Absolute Daily Return Rank	
$Score_{i,t}$	12.8722^{***}	19.9989^{***}
	(0.5578)	(1.7311)
$\mathbf{1}\{r_{i,t} < 0\}$	38.8560^{***}	60.4367^{**}
	(8.0687)	(25.5506)
$Score_{i,t} \times \mathbf{1}\{r_{i,t} < 0\}$	-1.9274^{***}	-6.9353^{***}
	(0.7288)	(2.2521)
Day FE	Yes	Yes

Table 8 reports the estimates. The coefficient on $Score_{i,t}$ is positive and highly significant for both Robinhood and TAQ, indicating that buying pressure is strongly increasing in extremity even within the top-mover set. The key distinction arises for top losers. In TAQ, the interaction term is negative and statistically significant, implying that the loser–winner gap becomes larger for the most extreme movers. In contrast, the corresponding interaction is weakly positive (overnight ranking) or much smaller in magnitude (daily ranking)

for Robinhood, consistent with the notion that the platform's mixed-signed Top Movers list attenuates the usual "gainer tilt" of retail attention. Interestingly these results align with the intraday observations of Ardia et al. (2023, p. 14), who find that Robinhood users are "particularly fast at opening positions in stocks that exhibit large negative price movements". This first exercise therefore suggests that the Top Movers List could be a powerful *driver* of retail attention and trading. Indeed, its design—ranking by absolute price movements—generates a distinctive pattern of buying responses to top gainers and top losers, supporting the hypothesis that platform interfaces and information salience can shape retail trading behavior.

6.1.2. Market Cap Threshold of Top Movers List

The rank-based evidence in the preceding subsection suggests that the Top Movers list operates as a salience channel that reallocates retail attention. A natural next question is whether this salience is merely correlated with Robinhood buying or whether the interface itself exerts a causal effect. A direct regression discontinuity design based on the boundary between the 20th and 21st mover would be ideal, but in practice the intraday ranking depends on a proprietary algorithm and is time-varying across the day[3], so that any proxy for the cutoff rank would introduce substantial measurement error. I therefore follow Barber et al. (2022) and exploit a sharp eligibility threshold embedded in the platform rules: stocks must exceed $300 million in market capitalization to be displayed in the Top Movers list.

The identifying variation comes from comparing stocks whose market capitalization at the market open lies just below versus just above the $300 million cutoff. In a small neighborhood around the threshold, stocks are similar in size, but only those above the cutoff are eligible to be displayed on the list. If the interface does not affect trading, then conditional on a smooth relation between size and trading activity, Robinhood investor demand should not exhibit a discontinuity at the cutoff. Conversely, if the interface *manufactures attention*, then eligibility should generate a discrete increase in buy-side participation. To implement the design, I restrict attention to "top mover" stocks within a bandwidth around the cutoff. For each day t, the treated sample consists of stocks with market cap in $(300, 300 + b]$ (in millions) that rank in the top 20 by absolute overnight return among stocks with market cap above $300 million. The control sample consists of stocks with market cap in $[300 - b, 300]$ whose absolute overnight return realizations are comparable to those of the treated observations[4]. I report results for bandwidths $b \in \{50, 75, 100, 125\}$ million, mirroring Barber et al. (2022).

[3] https://cdn.robinhood.com/disclosures/WebDisclosures.pdf.

[4] Specifically, for each stock in the treatment group, I search a matched stock that has the closest absolute return distance with the treated stock among all stocks with market cap $\in [300 - b, 300]$. I also further restrict the matching procedure by allowing only control stocks for which $|R^{\text{treatment}}_{\text{overnight},t}|/|R^{\text{control}}_{\text{overnight},t}| \in [0.5, 2]$.

I estimate the pooled sharp RD specification expressed in Equation 6.2.

$$
\begin{aligned}
userchg_{i,t}^{\text{intra}} = \beta_0 + \beta_1 \mathbf{1}\{size_{i,t} > 300\} + \sum_{n=1}^{N} \beta_2^{(n)} (size_{i,t} - 300)^n \\
+ \sum_{n=1}^{N} \beta_3^{(n)} (size_{i,t} - 300)^n \cdot \mathbf{1}\{size_{i,t} > 300\} + \varepsilon_{i,t},
\end{aligned}
\tag{6.2}
$$

Where $userchg_{i,t}^{\text{intra}}$ is Robinhood intraday user change on day t and $size_{i,t}$ is market capitalization at the market open of day t, both expressed in millions. The polynomial order N is set to 1, 2, or 3. Standard errors are clustered by day.

Figure 8. Robinhood Intraday User Change around the $300 Million Market-Cap Cutoff. This figure shows a binned scatterplot for the Robinhood (RH) intraday user change against market cap for stocks included in the regression discontinuity analysis as described in Barber et al. (2022). The sample consists of Top Mover stocks used in the RD analysis within a $50 million bandwidth around the $300 million cutoff. Solid curves are fitted lines estimated separately on each side of the cutoff under linear (Panel A), quadratic (Panel B), and cubic (Panel C) specifications. Shaded areas denote 90% confidence intervals.

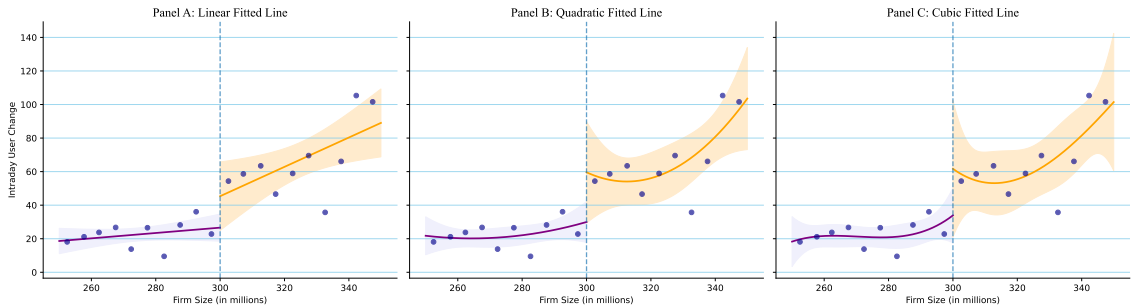


Figure 8 provides graphical evidence for the baseline bandwidth ($b = 50$) under linear, quadratic, and cubic fitted functions estimated separately on each side of the cutoff. The fitted relationship between intraday user change and market cap evolves smoothly away from the cutoff, but exhibits a discrete upward shift at $300 million, consistent with an interface-driven increase in the salience of eligible top movers.

Table 9 reports the corresponding regression estimates for the various bandwidths and polynomial orders, as well as the placebo cutoff results. Across specifications, the coefficient on the above-cutoff indicator β_1 is positive, aligned with the hypothesis that the platform's eligibility rule amplifies Robinhood buying responses to extreme movers. However, the statistical significance of the discontinuity is sensitive to the choice of bandwidth and polynomial order and difficult to establish, with the most robust evidence emerging for the linear specification. On average, the point estimates suggest that crossing the $300 million threshold increases intraday user change by around 30 users, which is economically meaningful given that the average intraday user change in the sample is around 2.75 users. As a falsification test, I additionally implement a placebo cutoff at $250

Table 9. Regression Discontinuity in Robinhood Intraday User Change for Top Mover Stocks

This table estimates a sharp regression discontinuity specification around the $300 million market-cap cutoff for inclusion in Robinhood's Top Movers list. The dependent variable is Robinhood intraday user change on day t. The key regressor is an indicator that equals one if market cap at the open of day t exceeds $300 million. Polynomial functions of the centered running variable ($size_{i,t} - 300$) are included as controls, with order $N = 1, 2, 3$. Panel bandwidths vary from $50 million to $125 million. All specifications include day-level clustering of standard errors. Standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Robinhood Intraday User Change		
	(1)	(2)	(3)
Panel A: Sample Bandwidth = $50 mil.			
Larger than 300 m	18.75	115.8*	27.773
	(13.995)	(29.632)	(24.903)
Polynomial order, N	1	2	3
Observations	2,811	2,811	2,811
R^2	0.027	0.027	0.027
Panel B: Sample Bandwidth = $75 mil.			
Larger than 300 m	29.25***	19.01	21.49
	(11.184)	(15.063)	(19.533)
Polynomial order, N	1	2	3
Observations	5,058	5,058	5,058
R^2	0.032	0.032	0.033
Panel C: Sample Bandwidth = $100 mil.			
Larger than 300 m	29.683***	19.07	21.95
	(9.597)	(13.602)	(16.943)
Polynomial order, N	1	2	3
Observations	7,834	7,834	7,834
R^2	0.039	0.039	0.039
Panel D: Sample Bandwidth = $125 mil.			
Larger than 300 m	30.91***	18.75	21.02
	(8.403)	(12.888)	(15.582)
Polynomial order, N	1	2	3
Observations	11,087	11,087	11,087
R^2	0.047	0.047	0.047
Panel E: Placebo Cutoff at $250 mil. & Sample Bandwidth = $50 mil.			
Larger than 250 m	0.00***	0.00***	0.00***
	(0.000)	(0.000)	(0.000)
Polynomial order, N	1	2	3
Observations	3,542	3,542	3,542
R^2	0.015	0.015	0.015

million; consistent with the absence of any display threshold at that value, the placebo discontinuity is statistically indistinguishable from zero across all specifications.

In other economic settings, Chetty et al. (2009) show that *information salience* can materially shape individual behavior. The evidence above is consistent with that view. Robinhood's Top Movers List constitutes a salient and cognitively inexpensive signal to process (Kahneman, 1973), and thus likely increases the probability that the stocks it displays enter retail investors' consideration sets. In this sense, the interface can *manufacture attention* by heightening the salience of selected tickers and, in turn, increasing retail trading activity in those stocks.

6.2. The Attention Transmission Mechanism

The preceding section established that Robinhood's Top Movers interface is a salient and behaviorally consequential signal: within the displayed set, Robinhood users respond approximately symmetrically to extreme gainers and losers, and trading around the $300 million eligibility threshold indicates that platform visibility can generate discrete increases in buy-side participation. This section goes one step further and asks whether platform-induced salience also propagates *outside* the app in the form of heightened external information acquisition. If the Top Movers list expands the set of stocks that retail investors actively consider, then it should not only affect trading, but also increase the likelihood that investors search for those stocks. Abnormal Google search volume $asvi_{i,t}$, the central metric of this thesis, provides a natural way to test this prediction. If the discontinuity in Robinhood trading at the $300 million cutoff is driven by the platform's ability to manufacture attention, then we should also observe a discontinuity in $asvi_{i,t}$ at the same threshold, reflecting that the platform's salience induces investors to seek information about the eligible stocks. It should nevertheless be pointed out that since Robinhood's user base is only a subset of the overall retail investor population, discontinuity in $asvi_{i,t}$ at the cutoff is not guaranteed, as the platform's influence on search attention is diluted by the presence of other non-Robinhood investors who do not respond to the Top Movers list. Hence, the presence of a discontinuity in $asvi_{i,t}$ would provide particularly strong evidence for the attention manufacturing hypothesis.

6.2.1. Estimation & Results

To operationalize this idea, I re-estimate the market-cap regression discontinuity design of Barber et al. (2022) but replace intraday user change with abnormal Google search intensity $asvi_{i,t}$ as the dependent variable (Da et al., 2011, 2025). The running variable is market capitalization at the market open, and the treatment indicator is eligibility for display on the Top Movers list, i.e. $\mathbf{1}\{size_{i,t} > 300\}$.

Specifically, on the same matched top-mover sample as in the previous RD analysis, I

estimate Equation 6.3.

$$
\begin{aligned}
asvi_{i,t} = \beta_0 + \beta_1 \mathbf{1}\{size_{i,t} > 300\} + \sum_{n=1}^{N} \beta_2^{(n)} (size_{i,t} - 300)^n \\
+ \sum_{n=1}^{N} \beta_3^{(n)} (size_{i,t} - 300)^n \cdot \mathbf{1}\{size_{i,t} > 300\} + \varepsilon_{i,t},
\end{aligned}
\tag{6.3}
$$

Where $size_{i,t}$ is market capitalization in millions at the market open of day t, and $N \in \{1, 2, 3\}$ denotes linear, quadratic, and cubic specifications. Standard errors are clustered by day, reflecting that the treatment assignment and the construction of the local top-mover set are day-specific.

Figure 9. Robinhood Abnormal Search Volume Index (ASVI) around the $300 Million Market-Cap Cutoff. This figure shows a binned scatterplot for the Robinhood (RH) abnormal search volume index $asvi_{i,t}$ against market cap for stocks. The sample consists of Top Mover stocks used in the RD analysis within a $50 million bandwidth around the $300 million cutoff. Solid curves are fitted lines estimated separately on each side of the cutoff under linear (Panel A), quadratic (Panel B), and cubic (Panel C) specifications. Shaded areas denote 90% confidence intervals.

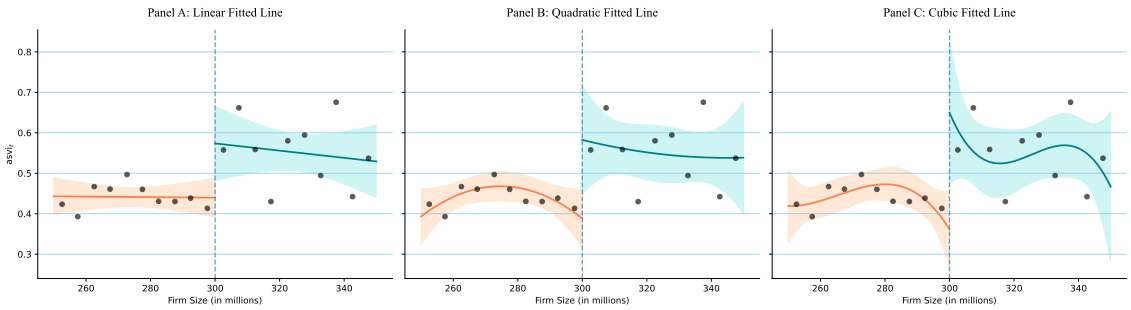


Figure 9 plots binned averages of $asvi_{i,t}$ against market capitalization within the baseline bandwidth of $50 million around the threshold, together with fitted lines estimated separately on each side of the cutoff. Across polynomial orders, the fitted relationship between abnormal search intensity and market cap is smooth away from the threshold but exhibits a visible upward shift at $300 million. This graphical discontinuity indicates that, conditional on being a top mover near the size cutoff, eligibility for the Top Movers display is associated with higher contemporaneous Google search attention.

Table 10 complements quantitatively Figure 9 and reports the corresponding RD estimates. The treatment effect $\hat{\beta}_1$ is positive and statistically significant across all bandwidths and polynomial controls. In the baseline window ($50 million), the estimated discontinuity ranges from 0.134 to 0.288 depending on the polynomial order. Because $asvi_{i,t}$ is defined as a log transformation, these magnitudes correspond to economically non-trivial increases in abnormal search activity: a coefficient of 0.134 implies roughly a 14% increase in $1 + svi/\overline{svi}$ at the cutoff, while a coefficient of 0.288 implies an increase

Table 10. Regression Discontinuity in Abnormal SVI for Top Mover Stocks

This table estimates a sharp regression discontinuity specification around the $300 million market-cap cutoff for inclusion in Robinhood's Top Movers list. The dependent variable is abnormal search volume index $asvi_{i,t}$ on day t. The key regressor is an indicator that equals one if market cap at the open of day t exceeds $300 million. Polynomial functions of the centered running variable $(size_{i,t} - 300)$ are included as controls, with order $N = 1, 2, 3$. Panel bandwidths vary from $50 million to $125 million. All specifications include day-level clustering of standard errors. Standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	$asvi_{i,t}$		
	(1)	(2)	(3)
Panel A: Sample Bandwidth = $50 mil.			
Larger than 300 m	0.134**	0.194**	0.288**
	(0.064)	(0.095)	(0.126)
Polynomial order, N	1	2	3
Observations	3,066	3,066	3,066
R^2	0.004	0.005	0.006
Panel B: Sample Bandwidth = $75 mil.			
Larger than 300 m	0.159***	0.170**	0.218**
	(0.051)	(0.076)	(0.102)
Polynomial order, N	1	2	3
Observations	5,488	5,488	5,488
R^2	0.003	0.003	0.004
Panel C: Sample Bandwidth = $100 mil.			
Larger than 300 m	0.170***	0.148**	0.223***
	(0.043)	(0.066)	(0.086)
Polynomial order, N	1	2	3
Observations	8,093	8,093	8,093
R^2	0.003	0.003	0.003
Panel D: Sample Bandwidth = $125 mil.			
Larger than 300 m	0.118***	0.183***	0.144*
	(0.038)	(0.060)	(0.077)
Polynomial order, N	1	2	3
Observations	11,023	11,023	11,023
R^2	0.002	0.002	0.002
Panel E: Placebo Cutoff at $250 mil. & Sample Bandwidth = $50 mil.			
Larger than 250 m	0.00***	0.00***	0.00***
	(0.000)	(0.000)	(0.000)
Polynomial order, N	1	2	3
Observations	3,353	3,353	3,353
R^2	0.001	0.001	0.001

of about 33%. The effect remains positive and precisely estimated in wider bandwidths, suggesting that the result is not driven by a narrow local anomaly. Finally, the placebo cutoff at $250 million yields no comparable jump, consistent with the absence of any display threshold at that value.

6.2.2. Origins & Confirmations of Attention

Google search is often regarded as a relatively direct measure of investor attention because it reflects an active information-acquisition decision rather than passive exposure (Da et al., 2011). Yet the present evidence indicates that search itself is subject to a prior selection stage. Retail investors face an enormous universe of potential securities, so some upstream mechanism must first determine which stocks enter the set of candidates deemed worth investigating, much as in the buying decision itself. The Top Movers list is precisely such a mechanism. By curating a small set of extreme movers and displaying them prominently within the trading interface, the platform lowers the effective cost of attention allocation and increases the likelihood that these stocks enter retail investors' consideration sets (Barber & Odean, 2008; Kahneman, 1973). Once a stock has entered that set, search naturally follows as a complementary action through which investors gather contextual information, read news, and verify recent price movements.

From this perspective, abnormal Google search intensity is best interpreted not as an autonomous driver of retail trading, but rather as a confirmatory manifestation of salience, a conclusion that is consistent with the timing evidence in chapter 5. The regression discontinuity design isolates a purely interface-driven shift in visibility, and the fact that $asvi_{i,t}$ increases at the eligibility threshold implies that search attention is, at least in part, a *subsequent* outcome of the platform's information architecture. Taken together, these findings suggest a useful classification of signals within the attention-trading literature. *Origination signals* are signals that directly trigger attention and trading, such as inclusion in the Top Movers list or extreme returns. *Confirmation signals*, by contrast, are signals that reveal whether this temporary salience has in fact translated into investor attention. In this framework, Google search is most plausibly understood as a confirmation signal rather than an origination signal.

6.2.3. Confirmation Signals and the Intensity of Herding Episodes

The distinction between origination and confirmation signals also yields a natural empirical prediction at the stock-day level. If an origination signal brings a stock into retail investors' consideration sets and thereby triggers a herding episode, then the magnitude of that episode should be stronger when the same stock-day also exhibits a confirmation signal in the form of abnormally high Google search intensity. Intuitively, once a stock has become salient enough to trigger herding, the presence of concurrent search activity indicates that

investors have indeed been hooked by the origination signals. In that case, one should expect more intense net inflows than during observationally similar herding episodes without a search spike. To evaluate this hypothesis, I estimate the interaction specification in Equation 6.4. The dependent variable is the daily Robinhood user change $userchg_{i,t}$, the indicator $rh_herd_{i,t}$ identifies stock-days classified as herding episodes, and $asvi_{i,t}$ measures contemporaneous abnormal Google search intensity.[5] The coefficient of interest is the interaction term between abnormal search intensity and the herding indicator.

$$\begin{aligned} userchg_{i,t} = \alpha_i + \delta_t + \beta_1 asvi_{i,t} + \beta_2 rh_herd_{i,t} \\ + \beta_3 \left(asvi_{i,t} \times rh_herd_{i,t} \right) + \theta \mathbf{X}_{i,\tau} + \varepsilon_{i,t} \end{aligned} \tag{6.4}$$

In Equation 6.4, α_i and δ_t denote stock and date fixed effects, respectively, and $\mathbf{X}_{i,\tau}$ contains the same controls used in previous regressions to the exception that some variables are not lagged anymore to reflect the contemporaneous nature of the specification. Specifically, $\mathbf{X}_{i,\tau}$ includes: $top_20_{i,t-1}$, $ab_vol_{i,t}$, $userchg_{i,t-1}$, $num_news_{i,t}$, $num_news_relevant_{i,t}$, $ess_{i,t}$, $css_{i,t}$, and $anl_chg_{i,t}$. Standard errors are two-way clustered by stock and date.

The coefficient β_1 captures the association between abnormal search attention and Robinhood user flows on non-herding stock-days. The coefficient β_2 measures the average increment in user flows during herding episodes, conditional on the included controls and fixed effects. Most importantly, β_3 tests whether the attention–flow relation is steeper during herding episodes than outside them.

Table 11 reports the estimates. The coefficient on $rh_herd_{i,t}$ is large and positive, indicating that user inflows increase sharply on stock-days classified as herding episodes. More importantly, the interaction term between $asvi_{i,t}$ and $rh_herd_{i,t}$ is positive, economically large, and precisely estimated. The point estimate implies that the association between abnormal search intensity and Robinhood user flows is substantially stronger during herding episodes than outside them. Quantitatively, the slope of $userchg_{i,t}$ with respect to $asvi_{i,t}$ equals $\hat{\beta}_1 = 1.1945$ on non-herding stock-days, but rises to $\hat{\beta}_1 + \hat{\beta}_3 = 466.4145$ when $rh_herd_{i,t} = 1$. Thus, conditional on the same stock and date effects and the same information set captured by the controls, abnormal search intensity is associated with a much steeper increase in user inflows precisely when retail trading is already concentrated in a herding state.

Overall, Table 11 supports our postulate. Origination signals such as extreme returns or platform visibility may place a stock into the consideration sets of retail investors and thereby trigger the onset of herding. Conditional on that onset, however, the intensity of

[5] The use of contemporaneous $asvi_{i,t}$ is deliberate. The purpose of this exercise is not to study whether Google search predicts future herding, but whether heightened search activity coincides with, and is associated with a stronger magnitude of, herding episodes once they occur. In the terminology developed above, the specification therefore targets the role of Google search as a *confirmation signal*.

Table 11. Interaction between Abnormal Search Attention and Herding Episodes

This table estimates Equation 6.4. The dependent variable is daily Robinhood user change $userchg_{i,t}$. $asvi_{i,t}$ is contemporaneous abnormal Google search intensity. $rh_herd_{i,t}$ is an indicator equal to one if stock i is in a Robinhood herding episode on day t. The coefficient on the interaction term $asvi_{i,t} \times rh_herd_{i,t}$ tests whether abnormal search attention is more strongly associated with user inflows during herding episodes than on other stock-days. All specifications include stock and date fixed effects. The table only reports the coefficients of interest. Standard errors, reported in parentheses, are two-way clustered by stock and date. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Robinhood User Change
$asvi_{i,t}$	1.1945^{***}
	(0.2408)
$rh_herd_{i,t}$	324.7800^{***}
	(72.0570)
$asvi_{i,t} \times rh_herd_{i,t}$	465.2200^{***}
	(94.5360)
Stock FE	Yes
Date FE	Yes
Observations	933,008
R^2	0.3778

the episode is significantly greater when the same stock-day also exhibits abnormally high search activity. In that sense, search (an origination signal) does not merely accompany salience; it identifies the subset of salient episodes in which retail attention appears to be more broadly engaged and, consequently, in which user inflows become especially pronounced. The following chapter will further explore the implications of these findings on returns dynamics.

VII. Effects of Attention-Induced Herding on Returns

[This chapter will analyze the asset pricing consequences of attention-induced herding, in particular by looking at return predictability and return reversals around herding episodes.]

VIII. Robustness & Limitations

[This chapter will conduct robustness checks as well as subsample analyses and discuss the several limitations of the study.]

IX. Conclusion

[The conclusion will be written last and will figure here.]

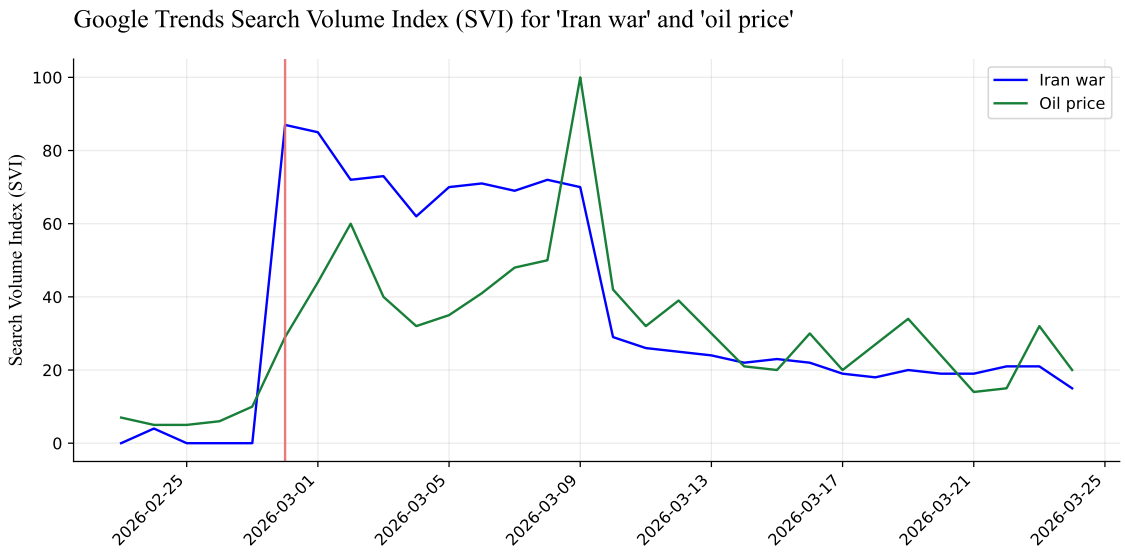
I. Additional Tables and Figures

A.1. Variable Construction Details

A.1.1. Google Trends and the Middle East

To illustrate how raw Google Trends data can be used, I plot in Figure 10 the search interest for the terms "Iran war" and "oil price" in Switzerland from February 23, 2026 to March 24, 2026. The plot shows that search interest for the two terms spiked on 28 February 2026, when the United States and Israel launched airstrikes against Iranian targets.

Figure 10. Google Trends search interest for the terms "Iran war" and "oil price" in Switzerland from February 23, 2026 to March 24, 2026. The red vertical line indicates the date of the airstrikes against Iranian targets on February 28, 2026.



A.1.2. RavenPack News Analytics

In this section, I provide additional details on the RavenPack News Analytics dataset, in particular the construction of the three sentiment variables. To build $ess_{i,t}$, $css_{i,t}$, and $anl_chg_{i,t}$, I follow the methodology outlined in Audrino et al. (2020) and repeated by De Giorgi et al. (2026). I aggregate sentiment scores as weighted sums of the sentiment scores of all news items related to stock i on day t, where the weights are given by the relevance scores of the news items.

Event Sentiment Scores (ESS)

The Event Sentiment Score (ESS) for stock i on day t is originally computed by RavenPack as a score in the interval $[-1, 1]$ with 2 decimal places that represents the news sentiment for a given entity by measuring various proxies sampled from the news—where -1 indicates extremely negative sentiment and 1 indicates extremely positive sentiment. The score is determined by systematically matching stories typically categorized by financial experts as having short-term positive or negative financial or economic impact. The final computation of the score depends on a particular proprietary algorithm that takes into account a significant number of factors. In this case, I simply aggregate the ESS scores of all news items related to stock i on day t as expressed in Equation A.1.

$$ess_{i,t} = \frac{1}{N_{i,t}^{ess}} \sum_{n=1}^{N_{i,t}^{ess}} ess_{n,t} \tag{A.1}$$

Where $N_{i,t}^{ess}$ denotes the number of news items related to stock i on day t, and $ess_{n,t}$ denotes the ESS associated with news item n for stock i on day t.

Composite Sentiment Scores (CSS)

The Composite Sentiment Score (CSS) is also a proprietary RavenPack score—$css_{i,t} \in [-1, 1]$ —stored with up to 2 decimal places. It proxies the news sentiment of a given story by combining various sentiment analysis techniques. The direction of the score is determined by looking at emotionally charged words and phrases and by matching stories typically rated by experts as having short-term positive or negative share price impact. However, since this signal does not account for the relevance of the news items, I follow Audrino et al. (2020) and compute a relevance-weighted CSS as expressed in Equation A.2.

$$css_{i,t} = \frac{\sum_{n=1}^{N_{i,t}^{css}} css_{n,t} \cdot rel_{n,t}}{100 \cdot N_{i,t}^{css}} \tag{A.2}$$

Where $N_{i,t}^{css}$ denotes the number of news items related to stock i on day t for which a CSS score is available, $css_{n,t}$ denotes the CSS associated with news item n for stock i on day t, and $rel_{n,t}$ denotes the relevance score of news item n for stock i on day t.

Analyst Recommendation Changes

This variable is a discrete score computed by RavenPack that captures changes in analyst recommendations. It takes the value of 1 if there is a positive change in the recommendation (e.g., from "hold" to "buy"), -1 if there is a negative change (e.g., from "buy" to "hold"), and 0 if there is no change. I compute the simple average of this score across all news items related to stock i on day t as expressed in Equation A.3.

$$anl_chg_{i,t} = \frac{\sum_{n=1}^{N_{i,t}^{anl_chg}} anl_chg_{n,t}}{100 \cdot N_{i,t}^{anl_chg}} \tag{A.3}$$

Where $N_{i,t}^{anl_chg}$ denotes the number of news items related to stock i on day t for which an analyst recommendation change score is available, and $anl_chg_{n,t}$ denotes the analyst recommendation change score associated with news item n for stock i on day t.

A.2. Differences in Buying and Selling User Flows Regression Results

This section reports the complete regression results for the determinants of Robinhood buy and sell flows. Importantly, the coefficents for the buy-side and sell-side regressions are comparable with caution, since the dependent variable in the sell-side regression is defined as the absolute magnitude of negative user changes, while the dependent variable in the buy-side regression is defined as the positive user changes. Hence, larger values in column (1) correspond to stronger sell-side outflows, while larger values in column (2) correspond to stronger buy-side inflows.

Table 12. Determinants of Robinhood Buy and Sell Flows

The table reports fixed-effects panel regressions estimated separately for buy-side and sell-side Robinhood user flows. In column (1), the dependent variable is $sellchg_{i,t}$, defined as the absolute magnitude of negative Robinhood user changes, i.e. $sellchg_{i,t} = -userchg_{i,t}$ for observations with $userchg_{i,t} < 0$. Hence, larger values in column (1) correspond to stronger sell-side outflows. In column (2), the dependent variable is $userchg_{i,t}$ restricted to observations with strictly positive user changes, so larger values correspond to stronger buy-side inflows. All regressions include stock and date fixed effects. $asvi_{i,t-1}$ is the lagged measure of abnormal search volume. $top_20_{i,t-1}$ is an indicator equal to one if the stock's absolute return was ranked in the top 20 on day $t-1$. $abvol_{i,t-1}$ is the logarithm of the ratio of stock market volume on day $t-1$ to the average volume from days $t-21$ to $t-2$. $userchg_{i,t-1}$ is the lagged change in Robinhood users. $num_news_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$. $num_news_relevant_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$ with RavenPack relevance score of at least 75. $ess_{i,t-1} \in [-1, 1]$ is the lagged Event Sentiment Score. $css_{i,t-1} \in [-1, 1]$ is the lagged relevance-weighted Composite Sentiment Score. $anl_chg_{i,t-1}$ is the lagged change in analyst recommendations. Robust standard errors, reported in parentheses, are two-way clustered by stock and date. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Dependent variable	
	$sellchg_{i,t}$ Sell-side ($userchg_{i,t} < 0$) (1)	$userchg_{i,t}$ Buy-side ($userchg_{i,t} > 0$) (2)
$top_20_{i,t-1}$	10.686***	35.639***
	(3.250)	(10.158)
$abvol_{i,t-1}$	10.259***	20.770***
	(2.241)	(7.259)
$userchg_{i,t-1}$	0.035	0.616***
	(0.038)	(0.047)
$asvi_{i,t-1}$	1.259***	−0.651
	(0.172)	(0.699)
$num_news_{i,t-1}$	−0.0002*	0.0003
	(0.0001)	(0.0003)
$num_news_relevant_{i,t-1}$	0.0008	−0.0025
	(0.0006)	(0.0017)
$ess_{i,t-1}$	−4.346*	−17.836***
	(2.361)	(6.480)
$css_{i,t-1}$	7.452	43.554***
	(4.559)	(16.762)
$anl_chg_{i,t-1}$	−46.976	9.678
	(111.840)	(388.190)
Observations	352,302	385,322
R^2	0.0058	0.3796
Entity FE	Yes	Yes
Date FE	Yes	Yes

A.3. Complete Event-Study Regression Results

This section reports the complete event-study regression results for the timing of the attention process. The main text only reports the coefficients for relative event times from $\tau = -5$ to $\tau = 5$ for brevity, but here I report the full set of coefficients for relative event times from $\tau = -10$ to $\tau = 10$.

Table 13. Abnormal Search Attention around Extreme Overnight Return Events

The table reports event-time regressions of abnormal Google search intensity around extreme overnight return events. Robust standard errors reported in parentheses are two-way clustered by event and calendar date. Column (1) reports the full event sample. Columns (2) and (3) split the sample according to whether the overnight return is positive or negative. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	$asvi_{i,t}$		
	(1) Overall	(2) Positive Returns	(3) Negative Returns
$\tau = -10$	−0.2100***	−0.2168***	−0.1527**
	(0.0398)	(0.0515)	(0.0766)
$\tau = -9$	−0.2036***	−0.1947***	−0.1623**
	(0.0352)	(0.0475)	(0.0666)
$\tau = -8$	−0.1949***	−0.1954***	−0.1474**
	(0.0313)	(0.0422)	(0.0589)
$\tau = -7$	−0.1868***	−0.1880***	−0.1632***
	(0.0283)	(0.0363)	(0.0526)
$\tau = -6$	−0.1720***	−0.1714***	−0.1406***
	(0.0249)	(0.0300)	(0.0459)
$\tau = -5$	−0.1559***	−0.1495***	−0.1382***
	(0.0198)	(0.0279)	(0.0353)
$\tau = -4$	−0.1273***	−0.1394***	−0.0937***
	(0.0163)	(0.0231)	(0.0286)
$\tau = -3$	−0.1098***	−0.1119***	−0.0883***
	(0.0128)	(0.0172)	(0.0224)
$\tau = -2$	−0.0939***	−0.1030***	−0.0818***
	(0.0122)	(0.0157)	(0.0187)
$\tau = 0$	0.1788***	0.1432***	0.1837***
	(0.0144)	(0.0184)	(0.0193)
$\tau = 1$	0.0463***	0.0397*	0.0252
	(0.0171)	(0.0225)	(0.0226)
$\tau = 2$	−0.0027	−0.0355	−0.0174
	(0.0192)	(0.0272)	(0.0278)
$\tau = 3$	−0.0157	−0.0395	−0.0429
	(0.0239)	(0.0315)	(0.0360)
$\tau = 4$	−0.0018	−0.0322	−0.0396
	(0.0236)	(0.0314)	(0.0430)
$\tau = 5$	0.0205	−0.0043	−0.0150
	(0.0278)	(0.0406)	(0.0486)
$\tau = 6$	0.0289	0.0003	−0.0160
	(0.0342)	(0.0459)	(0.0588)
$\tau = 7$	0.0452	0.0017	0.0047
	(0.0370)	(0.0534)	(0.0637)
$\tau = 8$	0.0425	−0.0125	−0.0051
	(0.0419)	(0.0562)	(0.0748)
$\tau = 9$	0.0627	0.0194	0.0108
	(0.0452)	(0.0630)	(0.0811)
$\tau = 10$	0.0731	0.0293	0.0140
	(0.0493)	(0.0670)	(0.0881)
Observations	131,187	78,603	72,849
R^2	0.0781	0.0843	0.0837
Event FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Reference Date	$\tau = -1$	$\tau = -1$	$\tau = -1$

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Declaration of Authorship

I hereby declare,

- that I have written this thesis independently,
- that I have written the thesis using only the aids specified in the index;
- that all parts of the thesis produced with the help of aids have been declared;
- that I have handled both input and output responsibly when using AI. I confirm that I have therefore only relied in public data or data released with consent and that I have checked, declared and comprehensibly referenced all results and/or other forms of AI assistance in the required form and that I am aware that I am responsible if incorrect content, violations of data protection law, copyright law or scientific misconduct (e.g. plagiarism) have also occurred unintentionally;
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